MORDELL-WEIL RANK BOUND FROM FREELY MOVABLE RATIONAL CURVES IN AN ELLIPTIC FIBRATION BASE

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

It is known that the rank of the Mordell-Weil group of a smooth elliptic K3 surface can only take values $0 \leq \mathrm{rkMW}(X/B) \leq 18$, and that each such value can be obtained [1, 2, 3]. These are the elliptic surfaces with section, singular fibers and $\kappa(X) = 0$.

Unlike the case of elliptic surfaces, very little is known about bounds of the rank of the Mordell-Weil group in higher dimensions. We prove an explicit bound on $\mathrm{rkMW}(X/B)$, for elliptic threefolds, with singular fibers and $\kappa(X) = 0$. For Calabi-Yau threefolds we prove the bound $\mathrm{rk\,MW}(X/B) \leq 28$, and for elliptic fourfolds we establish under natural assumptions the bound $\mathrm{rk\,MW}(X/B) \leq 38$. The precise statements are given in Theorem 6.1, Theorem 6.13 and Theorem 7.3.

We present two types of arguments with a common philosophy: the reduction to particular elliptic surfaces $Z \subset X$. We prove a natural inclusion of the Mordell-Weil groups $\mathrm{MW}(X/B) \subseteq \mathrm{MW}(Z/C)$ (Section 5). The two types of arguments provide different sufficient conditions for the existence of the surface Z. Each method leads to possible generalizations and applications in complementary directions.

For Calabi–Yau threefolds and fourfolds, the existence of a bound on $\mathrm{rkMW}(X/B)$ follows from the boundedness of families of elliptic threefolds [4] (see also [5]) and birational boundedness of fourfolds [6]. However, existing boundedness results for elliptic Calabi-Yau do not provide explicit numerical bounds on the Mordell–Weil rank and the boundedness of other elliptic varieties with $\kappa(X) = 0$ is proven under additional conditions, (for example, excluding some product type), [7, 8, 4, 5] and the recent [9].

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Our results provide explicit numerical Mordell–Weil bounds and they also apply in certain product type and non birationally bounded families.

At the same time, the highest Mordell–Weil rank currently known for elliptic Calabi-Yau threefolds is $\mathrm{rkMW}(X/B) = 10$ [10, 11]. Also, for Calabi-Yau manifolds, results from physics give the bound $\mathrm{rk\,MW}(X/B) \leq 28$ for threefolds and provide partial bounds for fourfolds under additional hypotheses [12, 13, 14], see Appendix A.

Our theorems prove (for Calabi-Yau threefolds) and go beyond (for Calabi-Yau fourfolds) the results obtained in physics [12, 13, 14].

Our methods apply beyond the Calabi–Yau setting. In particular, we establish explicit uniform Mordell–Weil bounds for elliptic threefolds of Kodaira dimension zero, even when the underlying varieties do not belong to birationally bounded families (see [8] and [7]). We improve the bounds to the known rank of 10 when the Mordell-Weil group has non trivial torsion and B is ruled.

The first method of reduction to a surface (Section 3) is via function field techniques, and the base curve of the surface is a general smooth movable rational curve without base points. This approach uses only elements of the classical minimal model theory for rational surfaces, without considering degenerate fibers, and avoids difficult results in higher dimensional birational geometry. It is formulated in the scheme language. In its current state this method does not provide an explicit effective bound for Calabi-Yau fourfolds.

The second method (Section 4) assumes that there are singular fibers, builds on a birational geometric approach, and on the properties of the Shioda pairing and its height. The base curve of the fibration is not necessarily movable nor rational, and it provides explicit effective bounds for Calabi-Yau fourfolds.

1.1. **Outline of the paper.** In Section 2 we collect the necessary background for our analysis, including results from the Minimal Model Problem applied to elliptic fibrations and properties of the Shioda pairing for rational sections. We also revisit Noether's formula (in all characteristics).

In Section 3, we prove the structure theorem 3.1 via field extensions. While philosophically similar to the procedure of the following Section 4, the proof presented in this section differs in two aspects. It uses that the Mordell-Weil rank of an elliptic curve over a field only increases on base change to a larger field, and that an elliptic curve over a function field of higher transcendence degree may sometimes be viewed as the generic fiber of an elliptic fibration on a smooth proper surface S_K, which is itself defined over some function field K. Then Noether's formula directly yields a bound for the Picard rank of this surface (and hence our desired elliptic Mordell-Weil rank), without considering degenerate fibers of the elliptic fibration, the Shioda pairing, or heights. The method in Section 3 is formulated in the scheme language. In its current state, it does not provide an explicit effective

bound for fourfolds, which we obtain, instead, by the method developed in Section 4.

In Section 4 we combine methods from birational geometry with fundamental properties of the Shioda map and its height-pairing to develop and prove the two structure theorems 4.15 and 4.17 for elliptic fibrations (not necessarily Calabi-Yau). The argument builds on the physics approach of [15, 12, 13, 14] and proves an injection from $\mathrm{MW}(X/B)$ to $\mathrm{MW}(Z/C)$ modulo torsion for a suitable surface Z (smooth, or birationally either a rational or K3 surface) obtained by restricting the elliptic fibration to a smooth movable curve C in the smooth locus of B. If C is not movable, other arguments are needed (see in particular Lemma 4.2).

In Section 5 we prove the inclusion of the whole Mordell-Weil groups, $\mathrm{MW}(X/B) \subseteq \mathrm{MW}(Z/C)$, given by the natural restriction of sections. We give sufficient conditions on the curves C for the inclusion to hold.

In Section 6, we apply Theorems 3.1 and 4.15 to prove a universal bound, formulated in Theorems 6.13, 6.1 and 6.3, on the Mordell-Weil rank of elliptic threefolds with $\kappa(X) = 0$, in particular Calabi-Yau. We describe the singularities on B in Corollary 6.11.

In Proposition 6.15 we prove explicit bounds for a broad class of threefold elliptic fibrations with $\kappa(X) < 0$.

Building on the results of Section 5 and 6 we describe the possible torsion subgroups of the Mordell-Weil group of elliptic threefolds in Theorem 6.14. In some cases these further constrain the rank of the free part.

In Section 7, we apply the structure theorems 4.15 and 4.17 to obtain an explicit bound for the Mordell-Weil rank of a broad class of elliptic fourfolds with $K_X \equiv 0$, in particular Calabi-Yau, Theorem 7.3. The proof requires a more delicate argument than the threefold case, because, while rational and very free curves of minimal degree are known to exist on rank-one Fano threefolds, it is a subtle problem to find those with minimal degree (with respect to the anticanonical divisor). We find that unlike for elliptic threefolds, the curves giving the best bound do not need to be movable nor are necessarily rational. A slightly modified argument proves explicit bounds for elliptic fourfolds fibrations with $\kappa(X) < 0$.

Our explicit bounds on $\mathrm{rkMW}(X/B)$ hold even when X does not belong to a (birational) bounded family, and it does not satisfy the hypotheses in [4], [8] and [7] which ensure boundedness of $\mathrm{rkMW}(X/B)$. The examples in Sections 6 and 7 illustrate that if the base B is not rationally connected, the elliptic fibration decomposes into an isotrivial component and a varying part. The Mordell–Weil rank depends on the non isotrivial component, and it remains uniformly bounded, Remark 6.6.

Our results suggest that the Mordell–Weil rank of an elliptic fibration should admit a linear bound depending only on the dimension of the total space, and we formulate this as a conjecture in Section 8.4.

Appendix A contains a review of related work from the physics of string theory. The original motivation of this project was in fact to prove the physics predictions for threefolds.

Acknowledgements

We thank Stefano Filipazzi, Seung-Joo Lee, Luca Martucci, Matthias Schütt, and Sho Tanimoto for discussions. TW especially thanks Seung-Joo Lee for collaboration on the related work [15, 12] and Luca Martucci and Nicolo Risso for collaboration on the related work [13]. TW thanks the University of Bologna for its hospitality. KHP wishes to thank VS and the University of Buffalo for hospitality during July-Aug 2024 when VS introduced him to the problems studied here that led to this paper. RM and VS wish to thank the organisers of the Summer Research Institute in Algebraic Geometry held in July 2025 at CSU, where we started discussing this together. VS thanks the UB Research Foundation for support to attend that event.

The work of TW is supported in part by Deutsche Forschungsgemeinschaft under Germany's Excellence Strategy EXC 2121 Quantum Universe 390833306, by Deutsche Forschungsgemeinschaft through a German-Israeli Project Cooperation (DIP) grant "Holography and the Swampland" and by Deutsche Forschungsgemeinschaft through the Collaborative Research Center 1624 "Higher Structures, Moduli Spaces and Integrability." The work of AG is supported in part by Prin 2022BTA242 "Geometry of algebraic structures: moduli, invariants, deformations" of Italian MUR. AG is a member of GNSAGA of INdAM.

2. Background on elliptic fibrations and their Mordell-Weil group

We work over the complex numbers; unless otherwise specified, all the varieties are assumed to be normal, projective and irreducible.

Definition 2.1. A complex variety X with canonical bundle ω_X is called Calabi-Yau if $\omega_X \simeq \mathcal{O}_X$ and $h^i(X, \mathcal{O}_X) = 0$ for $0 < i < \dim(X)$.

2.1. **Movable curves and effective divisors.** Let B be a $\mathbb{Q}$-factorial variety; let $B^{sm} \subset B$ be the non singular locus of B. Let $N_1(B)_{\mathbb{R}}$ be the set of numerical equivalence classes of real 1-cycles and $N^1(B)_{\mathbb{R}}$ the set of numerical equivalence classes of real divisors.

Definition 2.2.
(1) A curve $C \subset B$ is movable if and only if there is an irreducible subvariety $\mathcal{C} \subset B \times T$, with T irreducible, such that all fibers C_t, $t \in T$, of the projection $\mathcal{C} \to T$ are irreducible curves in B, $C = C_{t_0}$ for some $t_0 \in T$ and the projection $\mathcal{C} \to B$ has dense image.
(2) With notation as above, if, in addition, the intersection $\cap_{t \in T} C_t$ is empty, we say that the curve $C \subset B$ is freely movable.

(3) $\overline{\mathrm{NM}(B)} = \overline{\{\sum a_i\gamma_i,\ a_i \geq 0,\ \gamma_i \in\ N_1(B)_{\mathbb{R}},\ \gamma_i \text{ movable }\}}$.
(4) $\overline{\mathrm{Eff}(B)} = \overline{\{\sum a_i D_i,\ a_i \geq 0,\ D_i \in\ N^1(B)_{\mathbb{R}},\ D_i \text{ effective }\}}$.

Theorem 2.3. [16] *Let B be projective and irreducible. Then:*

$$\overline{\mathrm{NM}(B)} = \{\gamma \in N_1(B)_{\mathbb{R}} \text{ such that } D\cdot\gamma \geq 0,\ \forall\ D \in \overline{\mathrm{Eff}(B)}\}.$$

Definition 2.4. [See for example, [17, Section 2.4]] Let $f : \mathbb{P}^1 \to B$ be a non constant immersion such that $f(\mathbb{P}^1) \subset B^{sm} \subseteq B$.

$C = f(\mathbb{P}^1)$ is said to be free (very free) if f^*T_B is nef (f^*T_B is ample), or equivalently $N_{C/B}$ is the direct sum of non-negative (positive) summands.

In the following we denote by $\rho(B) := \mathrm{rk}\,\mathrm{Pic}(B)$.

2.2. **Elliptic fibrations.**

Definition 2.5. An elliptic fibration is a morphism $\pi : Y \to V$ whose fibers over a dense set in V are elliptic curves. The complement of this dense locus in V, the discriminant of the fibration, is denoted by $\Sigma_{Y/V}$.

Remark 2.6. For X a variety, we indicate by K_X the canonical divisor, interpreted as the first Chern class of the canonical bundle ω_X, and whenever a divisor Λ is associated with a line bundle, we call the latter $\mathcal{L}$.

Definition 2.7. Let $\tilde{\pi} : Y \to V$ be an elliptic fibration. A birationally equivalent elliptic fibration $\pi : X \to B$ is a fibration which makes the diagram

$$\begin{array}{ccc} Y & \overset{\phi}{\dashrightarrow} & X \\ {\scriptstyle\tilde{\pi}}\downarrow & & \downarrow{\scriptstyle\pi} \\ V & \overset{\psi}{\dashrightarrow} & B \end{array}$$

commutative, where ϕ and ψ are birational maps.

Definition/Theorem 2.8 (after [18])**.** Let $\tilde{\pi} : Y \to V$ be an elliptic fibration between smooth varieties. To the general point of each irreducible (reduced) codimension 1 component Σ_i of the discriminant $\Sigma_{Y/V}$, there is an associated $a_i \in \mathbb{Q}$. $\Lambda_{Y/V} \overset{def}{=} \sum_i a_i\Sigma_i$ is the discriminant $\mathbb{Q}$-divisor.

Theorem 2.9. *Let $Y \to V$ be an elliptic fibration between smooth varieties, Λ_V be the discriminant $\mathbb{Q}$-divisor. Then the following properties hold:*

(1) $\kappa(Y) = \kappa(V + \Lambda_V)$.
(2) If $\Lambda_V \neq 0$, *then* $h^1(Y, \mathcal{O}_Y) = h^1(V, \mathcal{O}_V)$.
(3) If either $\dim(Y) \leq 5$ *or* $\dim(Y) > 5$ *and klt flips exist and terminate in* $\dim Y - 1$, *there exists a birationally equivalent elliptic fibration* $\pi : X \to B$, X *with $\mathbb{Q}$-factorial terminal singularities,* (B, Λ) *with $\mathbb{Q}$-factorial klt singularities, such that* Λ *is supported on* $\Sigma_{X/B}$, *the discriminant of the fibration, and* $K_X \equiv \pi^*(K_B + \Lambda)$.
(4) If $\kappa(Y) \geq 0$ *and either* $\dim(Y) \leq 5$ *or* $\dim(Y) > 5$ *and klt flips exist and terminate in* $\dim Y - 1$, X *can be taken to be minimal.*

(5) *If* $\dim(Y) \leq 5$, π *can be assumed to be equidimensional over an open set* $U \subset B$ *with* $\operatorname{codim}(B \setminus U) \geq 3$.

Furthermore:

(6) *If* $\dim Y = 3$, *there are no multiple fibers,* $\kappa(X) = 0$ *and either* $\omega_X \simeq \mathcal{O}_X$ *or* $h^1(\mathcal{O}_X) \geq 1$, *then* B *is smooth.*
(7) *If* $\dim Y = 3$, $\omega_X \simeq \mathcal{O}_X$, $h^1(X, \mathcal{O}_X) = 0$, *and there are no multiple fibers, then* B *is either* $\mathbb{P}^2$ *or the blow up of the Hirzebruch surface* $\mathbb{F}_n$, $n \leq 12$.
(8) *If* $\kappa(Y) = 0$, $\Lambda \neq 0$ *and either* $\dim(Y) \leq 5$ *or* $\dim(Y) > 5$ *and klt flips exist and terminate in* $\dim Y - 1$, *then* B *is birationally a Mori fiber space (see Definition 2.11).*

Proof. (2), (6), (7) are proved in [19] and [20]; the other statements are found in [21], which generalizes to higher dimensions the results for threefolds in [19, 22]. □

Remark 2.10. Note that the Minimal Model Program is known to hold in general when $\dim Y \leq 4$ (Y is elliptically fibered and K_Y cannot be a big divisor).

Definition 2.11. A Mori fiber space is a dominant projective morphism $f : B' \to U$ such that

(a) B' is a normal projective variety with $\mathbb{Q}$-factorial terminal (klt) singularities,
(b) $f_*(\mathcal{O}_{B'}) = \mathcal{O}_U$ and $\dim B' > \dim U$,
(c) $\rho(B') - \rho(U) = 1$,
(d) $-K_{B'}$ is f-ample, that is the general fiber is a Fano variety.

The general fiber F also has $\mathbb{Q}$-factorial terminal (klt) singularities, since B' does. By the classical results on the Minimal Model Program for threefolds, see for example [23], we have the following more precise structure theorem:

Theorem 2.12. *Let* $\pi : X \to B$ *an elliptic fibration as in Theorem 2.9,* $\dim(X) \leq 5$, *and* $K_X = \pi^*(K_B + \Lambda)$. *Assume also that* $\Lambda \neq 0$. *Then there is a birational transformation* $\mu : B \dashrightarrow B'$, *where* B' *has the structure of a Mori fiber space.*

Furthermore, if B *has terminal singularities, the following possibilities can occur:*

(1) *Any component of a fiber of* f *is a smooth rational curve. The general fiber is a smooth rational curve.*
(2) *The general fiber of* f *is a smooth del Pezzo surface.*
(3) B' *is a Fano variety with* $\mathbb{Q}$*-factorial terminal singularities,* $\rho(B') = 1$.
(4) *If* $\dim B' = 4$, *the general fiber* F *is a Fano threefold with* $\mathbb{Q}$*-factorial terminal singularities.*

Definition 2.13. Let B be a projective variety. B is rationally connected if and only if for any two very general points in B there is an irreducible rational curve which contains both points.

Definition 2.14. Let B be a projective variety. A maximal rationally connected (MRC) fibration of B is a rational map $\phi : B \dashrightarrow Z$ such that the general fiber is rationally connected and the base Z is not uniruled.

A MRC as in Definition 2.14 exists and it is unique up to birational equivalence [24, 25].

2.3. **Noether's Formula.**

Proposition 2.15. *Let $\pi\colon Z \to C$, where Z is a smooth proper surface over an algebraically closed field of characteristic 0, C is a smooth curve of genus $g(C)$ and $\omega_Z = \pi^*(\omega_C \otimes \mathcal{L}_C)$, for some line bundle $\mathcal{L}_C$ of degree > 0. Then*

$$\rho = \operatorname{rk} \mathrm{NS}(Z) \leq 10 \deg \mathcal{L}_C + 2g(C). \tag{1}$$

In particular, if $\pi : X \to B$ is as in Theorem 2.9, $\mathcal{L}$ is a line bundle on B, $C \subset B$ a smooth projective curve and $Z = \pi^{-1}(C)$, so that $\deg \mathcal{L}_C = \mathcal{L} \cdot C > 0$, then

$$\rho = \operatorname{rk} \mathrm{NS}(Z) \leq 10\mathcal{L} \cdot C + 2g(C). \tag{2}$$

Proof. Let Z be a smooth proper surface over any algebraically closed field (of any characteristic). Recall from [26] that Noether's formula also applies in this case:

$$\chi_{\mathrm{top}}(Z) = 12\,\chi(\mathcal{O}_Z) - K_Z^2, \tag{3}$$

where $\chi(\mathcal{O}_Z) = 1 - h^1(Z, \mathcal{O}_Z) + p_g$ and $p_g = h^{2,0}(Z)$.

On the other hand, in characteristic zero, even if the field is not the complex numbers [27, Theorem 5.5]:

$$\chi_{\mathrm{top}}(Z) = \sum (-1)^i b_i(Z) = 2 - 4\,h^{0,1}(Z) + 2p_g + h^{1,1}(Z). \tag{4}$$

By combining (3) and (4) we have:

$$h^{1,1}(Z) = 10 - 8\,h^{0,1}(Z) + 10p_g - K_Z^2. \tag{5}$$

Now assume that Z admits an elliptic fibration

$$\pi\colon Z \to C,$$

where C is a smooth curve and $\omega_Z = \pi^*(\omega_C \otimes \mathcal{L}_C)$. By Riemann–Roch and using

$$p_g = h^0(Z, \omega_Z) = h^0(C, \omega_C \otimes \mathcal{L}_C)$$

one finds

$$\begin{aligned} p_g &= h^1(C, \omega_C \otimes \mathcal{L}) + 2g - 2 + \deg \mathcal{L}_C + 1 - g(C) \\ &= h^0(C, \mathcal{L}_C^{-1}) + g(C) - 1 + \deg \mathcal{L}_C = g(C) - 1 + \deg \mathcal{L}_C \end{aligned} \tag{6}$$

(since $h^0(C, \mathcal{L}_C^{-1}) = 0$, as $\mathcal{L}_C^{-1}$ has negative degree).

With (6) and observing that $K_Z^2 = 0$, we deduce from (5) that

$$\begin{aligned} h^{1,1}(Z) &= 10 + 10p_g - 8h^1(Z, \mathcal{O}_Z) = 10 + 10[g(C) - 1 + \deg \mathcal{L}_C] - 8g(C) \\ &= 10 \deg \mathcal{L}_C + 2g(C)\,. \end{aligned}$$

With $\operatorname{rk} \mathrm{NS}(Z) \leq h^{1,1}(Z)$ this yields (1). □

A slightly modified proof of Proposition 2.15 gives the following weaker bounds:

Proposition 2.16. *Let $\pi\colon Z \to C$, where Z is a smooth proper surface over an algebraically closed field of any characteristic, C is a smooth curve of genus $g(C)$ and $\omega_Z = \pi^*(\omega_C \otimes \mathcal{L}_C)$, for some line bundle $\mathcal{L}_C$ of degree > 0. Then*

$$\rho(Z) \leq b_2(Z) \leq 12 \deg \mathcal{L}_C + 4g(C) - 2$$

and

$$\rho(Z) \leq b_2(Z) \leq 12\mathcal{L} \cdot C + 4g(C) - 2\,,$$

respectively.

2.4. **The Shioda map and the Shioda pairing for elliptic fibrations with sections.**

From now on we will consider elliptic fibrations with a rational section.

Let $\mathrm{MW}(X/B)$ be the Mordell-Weil group of the elliptic fibration $\pi : X \to B$, defined as the group of rational sections of the fibration. Equivalently, we can write the Mordell-Weil group as $E(K)$ for $K = K(B)$ the function field of B. The zero section (the zero element of the group) will be denoted by S_0 (or, if the zero section is rational, by the closure of its image). In particular the existence of a zero section implies that there are no multiple fibers. Also, there is a canonical identification of groups between $\mathrm{Pic}^0(E_K)$ and the Mordell-Weil group $E(K)$.

Proposition 2.17. *Let $Y \to V$ and $X \to B$ be birationally equivalent fibrations as in Definition 2.7. Then* $\mathrm{MW}(Y/V) \simeq \mathrm{MW}(X/B)$.

Proof. For two birationally equivalent fibrations as in Definition 2.7, the generic elliptic fibers over the (common) function field of the two bases are isomorphic, and therefore have the same Mordell-Weil groups. □

Therefore, to derive bounds on the rank of the Mordell-Weil group, we can, without loss of generality, assume that X, B and $\pi : X \to B$ are as in Theorem 2.9.

Proposition 2.18. *If $C \subset B$ is a general smooth movable curve, and Z denotes the elliptic surface obtained by restricting the elliptic fibration of X to C, then* $\mathrm{MW}(X/B)_{\mathrm{tors}} \subseteq \mathrm{MW}(Z/C)_{\mathrm{tors}}$.

Proof. The locus where two different torsion sections coincide is contained in a proper divisor of B. A sufficiently general member of a movable family curve C is not contained in this locus. □

From now on, we assume that the rank of $\mathrm{MW}(X/B)$ is non-vanishing so that there are non-torsion rational sections, i.e. $\mathrm{rk}\,\mathrm{MW}(X/B) \neq 0$.

Definition 2.19. [28, 29] For an elliptic fibration $\pi : X \to B$ as above, if $\dim B \leq 3$, there is a Shioda map from the Mordell-Weil group to the Neron-Severi group $\mathrm{NS}(X) \otimes \mathbb{Q}$,

$$\sigma \,:\, \mathrm{MW}(X/B) \longrightarrow \mathrm{NS}(X) \otimes \mathbb{Q}\,,$$

which associates to a section S_a the class

$$\sigma(S_a) \;=\; S_a - S_0 - \pi^*\big(\pi_*((S_a - S_0)\cdot S_0)\big) + \sum_i \sum_A \ell^i_A(S_a)\, E^A_i. \tag{7}$$

Here

- S_a denotes the divisor (section) corresponding to the section a (or, if the section is rational, the closure of its image);
- S_0 is the zero section;
- E^A_i is a rationally fibered divisor such that $\pi(E^A_i)$ is an irreducible component $\Sigma_A \subseteq \Sigma_{X/B}$ and E^A_i is the i-th irreducible component of $\pi^{-1}(\Sigma_A)$;
- $\ell^i_A(S_a)$ are rational coefficients which measure how S_a meets the reducible fibers.

Remark 2.20. (a) The Shioda map is a group homomorphism.

(b) The term $\ell^i_A(S_a)$ contains as a factor the intersection multiplicity of S_a with the general fibre of the divisors E^A_i. In particular, if S_a intersects the singular fibres in the same component as the zero-section over the generic point of the discriminant, then all $\ell^i_A(S_a) = 0$.

(c) The restriction on $\dim B$ is not needed in Definition 2.19, though this is not explicitly done in the literature. This can be deduced from the Hard Lefschetz theorem and the Hodge index theorem (the pure Hodge structure $\mathrm{NS}(X) \otimes \mathbb{Q}$ carries a polarization, determined using Hodge theory).

(d) Theorem 2.9 (1) implies that there is a surjection

$$\mathrm{NS}(X) \otimes \mathbb{Q} \twoheadrightarrow \mathrm{Pic}(E_K) \otimes \mathbb{Q} = \mathrm{MW}(X/B) \otimes \mathbb{Q} \oplus \mathbb{Q},$$

and the kernel of the induced projection onto $\mathrm{MW}(X/B) \otimes \mathbb{Q}$ is a non-degenerate subspace for the polarization, because of the Hodge-Riemann bilinear relations (which is the Hodge index theorem, for $\mathrm{NS}(X)$), hence is a split surjection of rational vector spaces. This also accounts for the fact that the Shioda map is a group homomorphism, whose kernel is the torsion subgroup.

(e) Finally, the Shioda map is injective after tensoring with $\mathbb{Q}$.

Definition 2.21. For a section S_a we define the class

$$b_{S_a} := -\pi_*\big(\sigma(S_a)\cdot\sigma(S_a)\big) = \langle \sigma(S_a), \sigma(S_a)\rangle\,,$$

which is viewed as an element of $N^1(B)_{\mathbb{R}}$. This is the usual height pairing associated to the section, if B is a smooth projective curve, so that $N^1(B)_{\mathbb{R}} = \mathbb{R}$.

Proposition 2.22. *Let S_a be any (rational) section distinct from the zero section S_0. Then there exist an open set $\mathcal{V} \subset B$ and an integer $m \in \mathbb{N}$ such that the multiple mS_a (viewed in the Mordell-Weil group) intersects the singular fibres in the same component as S_0 over $\mathcal{V}$.*

Proof. Let $\mathcal{V} \subset B$ be the open set where S_a is regular. The statement follows from an extension of the proof of Lemma (1.5) in [30] (or standard results on the behaviour of sections under multiplication by integers). (See [15] for details.) Over $\mathcal{V}$ the multiple mS_a does not intersect the non-identity components of singular fibers for suitable m, and hence lies in the same component as S_0 there. □

Remark 2.23. We note that the smooth locus of any fibre of π has a group structure, which is a semidirect product of the subgroup given by the 0-component, and a finite abelian group (the group of connected components). Proposition 2.22 is a consequence.

Proposition 2.24. *Let S_a be any section distinct from the zero section S_0. Then there exists an integer $m \in \mathbb{N}$ such that*

$$m\sigma(S_a) \;=\; mS_a - S_0 - \pi^*\big(\pi_*((mS_a - S_0)\cdot S_0)\big). \tag{8}$$

Proof. The statement follows from Proposition 2.22 and Remark 2.20. □

Proposition 2.25. *Assume $\omega_X \simeq \pi^*(\omega_B \otimes \mathcal{L})$, with $\mathcal{L}$ a line bundle on B with associated divisor Λ. Then for all S_a, $-\pi_*(S_a \cdot S_a) = \Lambda$.*

Proof. The statement follows from an argument in [31, (7.30)]. □

Corollary 2.26. *Assume that $\omega_X \simeq \mathcal{O}_X$. Then for every section S_a we have*

$$-\pi_*\big(S_a \cdot S_a\big) \;=\; -K_B.$$

Proposition 2.27. *Assume $\omega_X \simeq \pi^*(\omega_B \otimes \mathcal{L})$, with $\mathcal{L}$ a line bundle on B with associated divisor Λ. Let S_a be any section distinct from the zero section S_0. Then there exists a positive integer m such that*

$$b_{S_a} = \frac{1}{m^2}\Big(2\Lambda + 2\pi_*\big((mS_a)\cdot S_0\big)\Big).$$

Furthermore $b_{S_a} \in \overline{\mathrm{Eff}(B)}$.

Proof. Let m be as in Proposition 2.22 so that mS_a meets the singular fibers in the S_0-component over generic points. The statement follows from Propositions 2.24 and 2.25. □

Corollary 2.28. *Let S_a be any section distinct from the zero section S_0. Assume that $\omega_X \simeq \mathcal{O}_X$. Then there exists a positive integer m such that*

$$b_{S_a} = \frac{1}{m^2}\Big(-2K_B + 2\pi_*\big((mS_a)\cdot S_0\big)\Big).$$

Furthermore $b_{S_a} \in \overline{\mathrm{Eff}(B)}$.

Finally, we recall the following:

Theorem 2.29 (Shioda-Tate-Wazir formula)**.** *Let $X \to B$ be as in Theorem 2.9 with $\Lambda \neq 0$. Then the rank of the Mordell-Weil group of X is bounded by the Neron-Severi group of X as follows:*

$$\mathrm{rk}(\mathrm{MW}(X/B)) \leq \mathrm{rk}(\mathrm{NS}(X)) - \mathrm{rk}(\mathrm{NS}(B)) - 1\,. \tag{9}$$

Proof. Terminal and klt singularities are rational and the statement follows from [32, Theorem 1.1], which extends [28, 33]. □

3. Bounds for the Mordell-Weil rank with Function Fields

In this section we present an arithmetic proof of the following structure theorem bounding the Mordell-Weil rank of elliptic fibrations:

Theorem 3.1. *Let $\pi : X \to B$ be a morphism between projective, normal complex algebraic varieties, such that*
(i) X is Cohen-Macaulay, and its singular locus has codimension at least 3;
(ii) π is an elliptic fibration with a section;
(iii) the canonical sheaf of X is trivial ($\omega_X \cong \mathcal{O}_X$).

Suppose that there is a freely movable smooth rational curve C in B with normal bundle $\mathcal{N}_C$ of degree d. Then the elliptic curve E_F over the function field $F = \mathbb{C}(B)$ has Mordell-Weil group $E_F(F)$ with

$$\mathrm{rk}E_F(F) \leq 18 + 10d = 10(-C)\cdot K_B - 2\,, \tag{10}$$

where d is the degree of the normal bundle of C_t for any $t \in T$.

Remark 3.2. From Definition 2.2 recall that C being a freely movable smooth rational curve in B has the following implications:

- There is an (irreducible) nonsingular complex variety T.
- There is a smooth morphism $\varphi : T \times \mathbb{P}^1_{\mathbb{C}} \to B$ with dense image, contained in the smooth locus of B.
- There is a $t_0 \in T$ such that $C_{t_0} = C$.
- Regarding this as a family of morphisms $\varphi_t : \mathbb{P}^1_{\mathbb{C}} \to B$, we may shrink T to assume that φ_t is an embedding for every t in T with image C_t.
- The family $\{C_t\}_{t\in T}$ has no base points (*i.e.*, for any $b \in B$ there exists $t \in T$ such that the corresponding rational curve C_t does not contain b).

Remark 3.3. Because of our hypothesis that the family of rational curves C_t has no base points, the normal bundle of any such C_t must be generated by its global sections, and so $d \geq 0$.

Remark 3.4. Movable curves are not necessarily base point free. However, the movable curves we consider in Section 6 are base point free.

3.1. **Proof of Theorem 3.1.** Recall that for an elliptic curve E over a field K, which has a given rational point $P \in E(K)$, the Mordell-Weil group $E(K)$ of K-rational points is naturally identified with the group $\mathrm{Pic}^0(E)$ of isomorphism classes of line bundles on E of degree 0, via the map

$$Q(\in E(K)) \mapsto \mathcal{O}_E(Q - P)(\in \mathrm{Pic}^0(E)).$$

Hence if $K \subset L$ is any extension of fields, then $\mathrm{Pic}^0(E) \to \mathrm{Pic}^0(E_L)$ is injective, since $E(K) \subset E_L(L)$ in an obvious way.

With the hypotheses and notation of Theorem 3.1, consider the "incidence variety"

$$Z = (T \times \mathbb{P}^1_{\mathbb{C}}) \times_B X := \{(t, u, x) \in T \times \mathbb{P}^1_{\mathbb{C}} \times X \mid \varphi(t, u) = \pi(x)\}.$$

The image $\pi(X_{sing})$ in B of the singular locus of X has codimension at least 2 in B, since X_{sing} has codimension at least 3, by assumption. Since φ is smooth, $\varphi^{-1}(\pi(X_{sing}))$ is also of codimension at least 2 in $T \times \mathbb{P}^1_{\mathbb{C}}$. Hence, if we replace T by a nonempty Zariski open subset, we may assume $\varphi(T \times \mathbb{P}^1_{\mathbb{C}})$ does not intersect $\pi(X_{sing})$. Now since φ is a smooth morphism, so is the morphism $f : Z \to X$ induced by the projection, and its image is disjoint from X_{sing}. Hence Z is nonsingular. It is also clear, since it is a pull-back, that the morphism $Z \to T \times \mathbb{P}^1_{\mathbb{C}}$ is an elliptic fibration with a cross section, since $X \to B$ has this structure.

The generic fibre of $f : Z \to T \times \mathbb{P}^1$ is $E_K = E \times_F K$, where $F = \mathbb{C}(B)$ is the function field of B, $K = \mathbb{C}(T \times \mathbb{P}^1_{\mathbb{C}})$ is the function field of $T \times \mathbb{P}^1_{\mathbb{C}}$, and $F \hookrightarrow K$ is the inclusion of fields induced by φ. Hence it suffices to prove the stated bounds on the Mordell-Weil rank for the elliptic curve E_K, since that rank can only increase after a base change of fields.

We next note that, if $L = \mathbb{C}(T)$ is the function field of T, we may consider the elliptic surface $Z_L \to \mathbb{P}^1_L$ obtained as the generic fiber of the composition

$$g : Z \to T \times \mathbb{P}^1_{\mathbb{C}} \to T$$

where the second map is the projection to T. Then $Z_L \to \mathbb{P}^1_L$ is a nonsingular projective surface defined over the field L, which has an elliptic fibration (the map to $\mathbb{P}^1_L$) with a section, and whose generic fiber is the *same* elliptic curve E_K.

We also note that for a Zariski open dense subset of $t \in T$, the fiber $Z_t = g^{-1}(t) = \pi^{-1}(C_t) \subset X$ is a nonsingular projective complex surface.

Consider an algebraic closure $\bar{L}$ of L, and the corresponding surface $Z_{\bar{L}}$. This surface comes equipped with an elliptic fibration

$$h : Z_{\bar{L}} \to \mathbb{P}^1_{\bar{L}}$$

with a cross section, and if $M = \bar{L}(\mathbb{P}^1)$ is the function field, then there are field extensions

$$K \hookrightarrow L \hookrightarrow \bar{L} \hookrightarrow M,$$

which allows us to identify the generic fiber of h with the elliptic curve

$$E_M = E_F \times_F M,$$

so that

$$E(F) \subset E(M)$$

is a subgroup. It clearly suffices to prove the bounds stated in Theorem 3.1 for this "new" elliptic curve E_M, which is the generic fiber of $h : Z_{\bar{L}} \to \mathbb{P}^1_{\bar{L}}$.

By the following lemma $H^1(Z_{\bar{L}}, \mathcal{O}_{Z_{\bar{L}}})$ vanishes. Hence the Picard group is discrete, and equal to its Neron-Severi group; by application of Theorem 2.29 we therefore have the estimate

$$\operatorname{rk} E(M) \leq \operatorname{rk} \mathrm{NS}(Z_{\bar{L}}) - 2\,.$$

The bound stated in Theorem 3.1 follows immediately from the lemma below.

Lemma 3.5. *Under the hypotheses of Theorem 3.1, and with the above notations, if d is the degree of the normal bundle of $C_t \subset B$, where $t \in T$, then $h^1(Z_{\bar{L}}, \mathcal{O}_{Z_{\bar{L}}}) = 0$ and*

$$\operatorname{rk}(\mathrm{NS}(Z_{\bar{L}})) \leq 20 + 10d\,.$$

Proof. We note that if $t \in T$, and $Z_t = \pi^{-1}(C_t) \subset X$ is such that Z_t is a nonsingular complex surface, with its induced elliptic fibration $\pi_t : Z_t \to \mathbb{P}^1_{\mathbb{C}}$, then (using adjunction on X, which has trivial canonical bundle) we see that Z_t has canonical line bundle

$$\omega_{Z_t} = \det\mathcal{N}_{Z_t} = \pi_t^*(\det\mathcal{N}_{C_t}),$$

where $\mathcal{N}_{Z_t}$ is the normal bundle of $Z_t \subset X$, and $\mathcal{N}_{C_t}$ is the normal bundle of C_t in B. Since $C_t \cong \mathbb{P}^1_{\mathbb{C}}$, we have that $\det\mathcal{N}_{C_t} \cong \mathcal{O}_{\mathbb{P}^1}(d)$ where d is the degree of the normal bundle of C_t. Similar reasoning shows that

$$\omega_{Z_{\bar{L}}} \cong h^*\mathcal{O}_{\mathbb{P}^1_{\bar{L}}}(d).$$

Using [34, Corollary 2.4(iv)] we see that $R^1h_*\mathcal{O}_{Z_{\bar{L}}} = \mathcal{O}_{\mathbb{P}^1}(-d-2)$. By the Leray spectral sequence for h, the vector space $H^1(Z_{\bar{L}}, \mathcal{O}_{Z_{\bar{L}}})$ is trapped between $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$ and $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-d-2)) = 0$; so it vanishes.

Hence the surface $Z_{\bar{L}}$ has geometric genus

$$p_g(Z_{\bar{L}}) = \dim H^0(Z_{\bar{L}}, \omega_{Z_{\bar{L}}}) = d + 1,$$

and

$$q(Z_{\bar{L}}) = \dim H^1(Z_{\bar{L}}, \mathcal{O}_{Z_{\bar{L}}}) = 0.$$

Noether's formula in Proposition 2.15 then implies that

$$\operatorname{rk}(\mathrm{NS}(Z_{\bar{L}})) \leq 12(1 + p_g) - 2 - 2p_g = 12(d+2) - 2 - 2(d+1) = 10d + 20.$$

□

The varieties X, B, T and the morphisms π and ϕ are defined over a countable algebraically closed subfield $k \subset \mathbb{C}$. The proof given above can be carried out over k instead of $\mathbb{C}$.

Remark 3.6. (i) Recall from Proposition 2.15 that Noether's formula is also valid for a smooth projective surface over an algebraically closed field, when interpreted suitably [26]; thus the resulting inequality on the rank of the Neron-Severi group is valid in characteristic 0 as given in Lemma 3.5 and a slightly weaker upper bound holds in any characteristic. Since $\bar{L}$ has characteristic 0, this can in fact be deduced from the case of complex surfaces, together with the assertion that for a surface S over an algebraically closed field K, and an extension of algebraically closed fields $K \subset L$, the Neron-Severi groups of S and of S_L are canonically isomorphic.

(ii) We recall that our Mordell-Weil group $E_F(F)$ corresponds to the group of *rational sections* of the elliptic fibration $\pi : X \to B$, since it clearly depends only on the generic fiber, viewed as an elliptic curve over the function field of the base variety.

4. Bounds for the Mordell-Weil rank via birational geometry

In the following we assume X as in Theorem 2.9, $K_X = \pi^*(K_B + \Lambda)$ and $\Lambda \neq 0$. The results in this section hold for any dimension, unless the dimension is specified. In particular, we will prove the structure theorems 4.15 and 4.17.

Proposition 4.1. *Let S_a be a section. Then $b_{S_a} \cdot C \geq 0$ for every curve class $C \in \overline{\mathrm{NM}(B)}$.*

Proof. By Proposition 2.27, $b_{S_a} \in \overline{\mathrm{Eff}(B)}$ and the cone $\overline{\mathrm{Eff}(B)}$ is dual to $\overline{\mathrm{NM}(B)}$, by Theorem 2.3. □

Lemma 4.2. *Let $S_a \in \mathrm{MW}(X/B)$, $C \subset B$ be a smooth curve contained in B^{sm}, the smooth locus of B, but such that $C \not\subset \Sigma_{X/B}$, and $Z \stackrel{def}{=} \pi^{-1}(C)$. Then*

$$b_{S_a} \cdot C = -\pi_{*|\widehat{Z}}\big(\sigma(S_{a|\widehat{Z}}) \cdot \sigma(S_{a|\widehat{Z}})\big) = \langle \sigma(S_{a|\widehat{Z}}), \sigma(S_{a|\widehat{Z}}) \rangle_{|_{\widehat{Z}}}$$

is well-defined and gives the height pairing $\langle , \rangle_{|_{\widehat{Z}}}$ for $S_a|_{\widehat{Z}} \in \mathrm{MW}(\widehat{Z}/C)$, where $\tau : \widehat{Z} \to Z$ is a minimal resolution of Z relative to the morphism $Z \to C$.

Proof. Recall that $b_{S_a} = -\pi_*\big(\sigma(S_a) \cdot \sigma(S_a)\big) = \langle \sigma(S_a), \sigma(S_a) \rangle$. By Proposition 2.22 there exists a positive integer m such that mS_a intersects the singular fibers, possibly after restricting to an open neighborhood, in the same component as S_0 and then $b_{S_a} = \frac{1}{m^2}\Big(2\Lambda + 2\pi_*\big((mS_a) \cdot S_0\big)\Big)$. There exists a Weierstrass model W_X of $X_{|\pi^{-1}(B^{sm})} \to B^{sm}$ and a Weierstrass model W_Z of $Z \to C$. Now the birational map between X and W_X is an isomorphism in an open set which includes $(mS_a) \cdot S_0$ for a suitable choice of m. Likewise, the birational maps between Z, $\widehat{Z}$ and W_Z are isomorphisms in an open set

of $((mS_a)\cdot S_0))_{|\pi^{-1}(C)}$. Then $\pi_*\big((mS_a)\cdot S_0)\big)\cdot C = (\pi_{|\widehat{Z}})_*\big((mS_a)_{|\widehat{Z}}\cdot S_{0|\widehat{Z}}\big)\Big)$, where $(mS_a)_{|\widehat{Z}}$ is also the addition in $\widehat{Z}$, by base change. We also have $K_{\widehat{Z}} = \pi^*_{|\widehat{Z}}(K_C + \Lambda\cdot C)$ and thus

$$\begin{aligned} b_{S_a}\cdot C &= \frac{1}{m^2}\Big(2\Lambda + 2\pi_*\big((mS_a)\cdot S_0\big)\Big)\cdot C \\ &= \frac{1}{m^2}\Big(2\Lambda\cdot C + 2(\pi_{|\widehat{Z}})_*\big((mS_a)_{|\widehat{Z}}\cdot S_{0|\widehat{Z}}\big)\Big) \end{aligned}$$

and

$$b_{S_a}\cdot C = -\pi_{*|\widehat{Z}}\big(\sigma(S_{a|\widehat{Z}})\cdot\sigma(S_{a|\widehat{Z}})\big) = \langle\sigma(S_{a|\widehat{Z}}),\sigma(S_{a|\widehat{Z}})\rangle_{|_{\widehat{Z}}}\,.$$

□

Remark 4.3. The compatibility of the height pairing with appropriate base changes, as recorded in Lemma 4.2, may be viewed as a special case of a formula in intersection theory, [35], Theorem 6.2 (a).

Notation 4.4. In the hypothesis of the previous Lemma 4.2, from now on we will write $\mathrm{MW}(Z/C)$ in place of $\mathrm{MW}(\widehat{Z}/C)$. In many cases indeed $Z = \widehat{Z}$.

Proposition 4.5. *Let $C\subset B^{sm}\subseteq B$ be a smooth curve, $C\not\subset\Sigma_{X/B}$ and $\pi^{-1}(C) = Z$. Assume that $b_{S_a}\cdot C\geq 0$ for every S_a and that $b_{S_a}\cdot C = 0$ if and only if $b_{S_a} = 0$, i.e. if S_a is torsion. Then*

(1) $S_a|_Z$ is torsion in $\mathrm{MW}(Z/C)\Longleftrightarrow S_a$ is torsion in $\mathrm{MW}(X/B)$,
(2) $S_a\neq S_b\in\mathrm{MW}(X/B)/\mathrm{MW}(X/B)_{\mathrm{tors}}\Longrightarrow$
$S_a|_Z\neq S_b|_Z\in\mathrm{MW}(Z/C)/\mathrm{MW}(Z/C)_{\mathrm{tors}}$.

Proof. Recall that $b_{S_a} = \langle S_a, S_a\rangle = 0$ if and only if S_a is torsion (see also Remark 2.20, (d)). The first statement follows then from the assumptions and Lemma 4.2. The same lemma and the Shioda homomorphism imply the second part. □

Theorem 4.6. *Let $C\subset B^{sm}\subseteq B$ be a smooth curve, $C\not\subset\Sigma_{X/B}$. Assume that $b_{S_a}\cdot C\geq 0$ for every S_a and that $b_{S_a}\cdot C = 0$ if and only if S_a is a torsion section. Then*

$$\mathrm{MW}(X/B)/(\mathrm{MW}(X/B))_{tor}\subseteq\mathrm{MW}(Z/C)/(\mathrm{MW}(Z/C))_{tor}$$

and

$$\mathrm{rk}\,\mathrm{MW}(X/B)\leq\mathrm{rk}\,\mathrm{MW}(Z/C)\,.$$

Proof. This follows from Proposition 4.5. □

Theorem 4.7. *Let $C\subset B^{sm}\subseteq B$ be a smooth curve such that $\pi^{-1}(C) = Z$ is a smooth surface. Assume that $b_{S_a}\cdot C\geq 0$ for every S_a and that $b_{S_a}\cdot C = 0$ if and only if S_a is a torsion section. Then*

$$\mathrm{rk}\,\mathrm{MW}(X/B)\leq\mathrm{rk}\,\mathrm{MW}(Z/C)\leq\mathrm{rk}\,\mathrm{NS}(Z) - 2\,.$$

Proof. The first inequality again follows from Proposition 4.5. The second inequality is an application of Theorem 2.29 to the smooth elliptic surface Z. □

Proposition 4.8. *Let $C \in \overline{\mathrm{NM}(B)}$ be a curve. Assume also that $\Lambda \cdot C > 0$. Then $b_{S_a} \cdot C \geq 0$ for every S_a, and $b_{S_a} \cdot C = 0$ if and only if S_a is torsion.*

Proof. Since $\Lambda \cdot C > 0$, Propositions 2.27 and 4.1 imply that $b_{S_a} \cdot C \geq 0$ and $b_{S_a} \cdot C = 0$ if and only if $b_{S_a} = 0$. □

Proposition 4.9. *Let $C \subset B$ be a curve. Assume also that $\Lambda \cdot C > 0$ and $\operatorname{rk} \mathrm{Pic}(B) = 1$. Then $b_{S_a} \cdot C \geq 0$ for every S_a, and $b_{S_a} \cdot C = 0$ if and only if $b_{S_a} = 0$, that is S_a is torsion.*

Proof. The hypotheses imply that $\overline{\mathrm{Eff}(B)}$ coincides with the closure in $\mathbb{R}$ of the ample cone. □

Corollary 4.10. *Let $C \subset B$ be a smooth curve. Assume that $\Lambda = -K_B$. Assume also that either $C \in \overline{\mathrm{NM}(B)}$ is rational or that $\operatorname{rk} \mathrm{Pic}(B) = 1$ with B Fano. Then $b_{S_a} \cdot C \geq 0$ for every S_a, and $b_{S_a} \cdot C = 0$ if and only if S_a is torsion.*

Proof. We only need to check the first case: if $C \in \overline{\mathrm{NM}(B)}$ is smooth and rational, the adjunction formula gives $-K_B \cdot C = 2 + \deg N_{C/B} > 0$. The statements follow from Propositions 4.8 and 4.9. □

Proposition 4.11. *Let $C \subset B^{sm} \subseteq B$ be a smooth rational curve. Assume that $b_{S_a} \cdot C \geq 0$ for every S_a, and $b_{S_a} \cdot C = 0$ if and only if $b_{S_a} = 0$, that is, S_a is torsion.*

(1) If in addition $\Lambda \cdot C = 2$, then $Z = \pi^{-1}(C)$ is (birationally) a K3 surface and

$$\operatorname{rk} \mathrm{MW}(X/B) \leq \operatorname{rk} \mathrm{MW}(Z/C) \leq 18.$$

(2) If in addition $\Lambda \cdot C = 1$, then $Z = \pi^{-1}(C)$ is (birationally) a rational elliptic surface and

$$\operatorname{rk} \mathrm{MW}(X/B) \leq \operatorname{rk} \mathrm{MW}(Z/C) \leq 8.$$

Proof. By construction, $Z = \pi^{-1}(C)$ is an elliptic surface with $12\Lambda \cdot C = 24$ or $12\Lambda \cdot C = 12$ singular fibers and is therefore birationally a K3 surface or a rational elliptic surface. The first part of the inequality follows from Theorem 4.6. The second part of the inequality follows from the known bound for Z a K3 surface or a rational elliptic surface (recalling Notation 4.4). □

Corollary 4.12. *Let $C \subset B^{sm} \subseteq B$ be a smooth rational curve with normal bundle $N_{C/B} \simeq \oplus^{\dim B - 1} \mathcal{O}_C$. Assume that $b_{S_a} \cdot C \geq 0$ for every S_a, and*

$b_{S_a} \cdot C = 0$ if and only if $b_{S_a} = 0$, that is, S_a is torsion. In addition assume that $\Lambda = -K_B$. Then $Z = \pi^{-1}(C)$ is birationally a K3 surface and

$$\operatorname{rk} \mathrm{MW}(X/B) \leq \operatorname{rk} \mathrm{MW}(Z/C) \leq 18\,. \tag{11}$$

Proof. By adjunction for the rational curve C, $\Lambda \cdot C = -K_B \cdot C = 2 + \deg N_{C/B} = 2$. The claim then follows from Proposition 4.11. □

Theorem 4.13. *Let $C \in \overline{\mathrm{NM}(B)}$ be a smooth curve such that $Z = \pi^{-1}(C)$ is smooth with $\Lambda \cdot C > 0$. Then*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq \operatorname{rk} \mathrm{MW}(Z/C) \leq \operatorname{rk} \mathrm{NS}(Z) - 2\,.$$

Proof. Proposition 4.8 implies that the hypotheses of Theorem 4.7 are satisfied. □

Theorem 4.14. *Let $C \subset B$ be a smooth curve such that $\pi^{-1}(C) = Z$ is smooth. Assume also that $\Lambda \cdot C > 0$ and $\operatorname{rk} \mathrm{Pic}(B) = 1$. Then*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq \operatorname{rk} \mathrm{MW}(Z/C) \leq \operatorname{rk} \mathrm{NS}(Z) - 2\,.$$

Proof. Proposition 4.9 implies that the hypotheses of Theorem 4.7 are satisfied. □

We then find a bound on $\operatorname{rk} \mathrm{MW}(X/B)$:

Theorem 4.15. *Assume there exists a curve $C \subset B^{sm}$ which is smooth, movable with $C \cdot \Lambda > 0$ and such that $Z = \pi^{-1}(C)$ is either smooth or (birationally) a K3 surface or a rational elliptic surface. Then*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10(C \cdot \Lambda) + 2g(C) - 2.$$

Proof. This follows from Theorem 4.13 together with Proposition 2.15 as well as Proposition 4.11. □

Remark 4.16. For X Calabi-Yau (in particular $K_X = 0$ and hence $\Lambda = -K_B \neq 0$) and C rational, the bound in Theorem (4.15) agrees with the physics bound

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10(-K_B \cdot C) - 2 \tag{12}$$

argued for in [13] for $\dim X = 3, 4$ by sharpening and generalising to Calabi-Yau 4-folds the conservative bounds of [12]. See also the discussion around (14) in the Introduction.

Theorem 4.17. *Let $C \subset B^{sm}$ be a smooth curve such that either $\pi^{-1}(C) = Z$ is a smooth surface or Z is (birationally) a K3 surface or a rational elliptic surface. Assume also that $C \cdot \Lambda > 0$ and $\operatorname{rk} \mathrm{Pic}(B) = 1$. Then*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10(C \cdot \Lambda) + 2g(C) - 2.$$

In particular if Z is birationally a K3 surface

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 18,$$

if Z is birationally a rational elliptic surface

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 8.$$

Proof. This follows from Theorem 4.14 together with Proposition 2.15 as well as Proposition 4.11. □

5. Reduction to elliptic surfaces

By combining the results of Section 3 with Proposition 2.18 and the results of Section 4 we obtain inclusions of the Mordell-Weil groups of sections of $X \to B$ to the Mordell-Weil group of a surface.

Given an algebraically closed sub-field k of $\mathbb{C}$ and a variety T over k, the k-embeddings of $L = k(T)$ in $\mathbb{C}$ give a subset V of the set $T(\mathbb{C})$ of (complex) points of T. A point lies in the complement of V if and only if it is a point of a proper k-subvariety of T. This set V is called the set of *very general* $\mathbb{C}$-points of T.

Theorem 5.1. *Given a complex elliptic fibration $\pi : X \to B$ with a section, and generic fiber E_K, where $K = \mathbb{C}(B)$, and a covering family of movable curves $\{C_t\}_{t\in T}$. Assume $MW(X/B) = E_K(K)$ is finitely generated.*

For a very general *$t \in T$, let $h : Z_t = \pi^{-1}(C_t) \to C_t$ be the restricted fibration with generic fiber E_F, where $F = \mathbb{C}(C_t)$.*

The natural restriction map $MW(X/B) = E_K(K) \to MW(Z_t/C_t) = E_F(F)$ is injective*.*

Proof. The varieties X, B, T and the morphisms π and ϕ are defined over a countable algebraically closed subfield $k \subset \mathbb{C}$. The proof of Theorem 3.1 can be carried out over k instead of $\mathbb{C}$. □

Remark 5.2. $MW(X/B) = E_K(K)$ is finitely generated, unless $E_K \cong E_0 \times_{\mathbb{C}} \mathrm{Spec}K$ is a base change of a complex elliptic curve, i.e., X is birational to $B \times E_0$ for a complex elliptic curve E_0. This is a special case of the Lang-Néron theorem; see for example [36, 37, 38] and [39, Théorème 2, with a detailed proof in Appendix 2].[1]

We also have:

Theorem 5.3. *Let $C \subset B^{sm}$ be a general smooth movable curve with $\Lambda \cdot C > 0$ and let $\pi^{-1}(C) = Z$. Then*

$$\mathrm{MW}(X/B) \subseteq \mathrm{MW}(Z/C)\,.$$

Proof. Proposition 4.8 and Theorem 4.6 prove that the natural restriction of sections is indeed an injection on the free part of the Mordell-Weil groups: $\mathrm{MW}(X/B)/\mathrm{MW}(X/B)_{\mathrm{tors}} \subseteq \mathrm{MW}(Z/C)/\mathrm{MW}(Z/C)_{\mathrm{tors}}$. If C is general and movable, we can assume that it is not contained in the divisor where the

[1] We thank Marc Hindry for correspondence on this point.

torsion sections coincide. Proposition 2.18 gives an injection on the torsion parts. □

Corollary 5.4. *Let $C \subset B^{sm}$ be a smooth curve general enough to be contained neither in the divisor where the torsion sections coincide, nor in the discriminant $\Sigma_{X/B}$. Assume that $\Lambda = -K_B$ and that B is Fano with $\rho(B) = 1$. Let $\pi^{-1}(C) = Z$. Then*

$$\mathrm{MW}(X/B) \subseteq \mathrm{MW}(Z/C)\,.$$

Remark 5.5. (1) In the above statements, Z does not need to be smooth.
(2) The curves C as in Corollary 5.4 will provide the bounds of the Theorems for fourfolds (Section 7).

6. Bounds for the Mordell-Weil rank of elliptic threefolds with $\kappa(X) = 0$.

When $\kappa(X) = 0$, then $\Lambda \equiv -K_B$. To obtain strong bounds on $\mathrm{rk}\,\mathrm{MW}(X/B)$ via the results of Sections 3 and 4 it is necessary to minimize $-K_B \cdot C + 2g(C) - 2$. The existence of rational curves $C \subset B$ of minimal degree with respect to the anticanonical K_B is a subtle problem in general.

We begin by proving the main result motivating this work, see also Remark 6.2.

6.1. Explicit bounds for the Mordell-Weil rank of elliptic Calabi-Yau threefolds.

Theorem 6.1. *Let $X \to B$ be an elliptic Calabi-Yau 3-fold as in Theorem 2.9, with $\Lambda \neq 0$. Then the Mordell-Weil group of X satisfies:*
(i) $\mathrm{rk}\,\mathrm{MW}(X/B) \leq 18$ *if* $B \neq \mathbb{P}^2$*;*
(ii) $\mathrm{rk}\,\mathrm{MW}(X/B) \leq 28$ *if* $B = \mathbb{P}^2$*.*

Proof. We present two different proofs, the first one, (a), follows the arithmetic proof of Section 3, the second, (b), the geometric proofs of Section 4.

(a) X has isolated (terminal) singularities. In case i), by Theorem 2.9 B is the blow up of a Hirzebruch surface and hence it has a rational fibration. There is therefore a morphism $\theta : B \to \mathbb{P}^1_{\mathbb{C}}$ whose general fiber is a smooth rational curve with the property that, for some Zariski open $T \subset \mathbb{P}^1_{\mathbb{C}}$, we have that $\theta^{-1}(T) \cong T \times \mathbb{P}^1_{\mathbb{C}}$. We may then take the family of rational curves C_t as in Theorem 3.1 to be the map $T \times \mathbb{P}^1_{\mathbb{C}}$ obtained from the fibres of θ. Here $d = 0$.

In case ii), take T to be an affine open set in the dual projective plane of lines in $B = \mathbb{P}^2_{\mathbb{C}}$, where we get a family of rational curves C_t as in Theorem 3.1 with the degree $d = 1$.

(b) In case i), the general rational fiber C of the (blow up of the) Hirzebruch surface B is movable and since $C \cdot \Lambda = C \cdot (-K_B) = 2$, $Z = \pi^{-1}(C)$ is (birationally) a K3 surface. Then by Theorem 4.15, $\mathrm{rk}\,\mathrm{MW}(X/B) \leq 18$.

In case ii) we can take C to be a general hyperplane class. Hence C is movable and $Z = \pi^{-1}(C)$ is smooth and $C \cdot (-K_B) = 3$. Theorem 4.15 implies $\operatorname{rk} \mathrm{MW}(X/B) \leq 28$.

□

Remark 6.2. For X a smooth Calabi-Yau threefold and B a ruled surface, the bound $\operatorname{rk} \mathrm{MW}(X/B) \leq 18$ appeared in the physics literature together with a list of possible torsion groups [14] [40]. See Appendix A for details.

When X is a Calabi-Yau threefold, or more generally a threefold with $K_X \simeq \mathcal{O}_X$ or $h^1(\mathcal{O}_X) = 1$, by Theorem 2.9, B is either a toric smooth surface or the blowup of a toric smooth surface. Then there exists a minimal generator of an extremal ray which represents a base point free movable rational curve, as toric nef divisors are base point free [41, Theorem 1.6]; then the proof of Theorem 6.3, along with Corollary 6.4 leads to the bounds described in the next subsection.

6.2. **Explicit bounds for the Mordell-Weil rank of all elliptic threefolds with $\kappa(X) = 0$.**

Theorem 6.3. *Let $X \to B$ be an elliptic threefold as in Theorem 2.9, with $\Lambda \neq 0$, $\kappa(X) = 0$. Then:*
(i) If $\operatorname{rk} \mathrm{Pic}(B) > 1$, then $\operatorname{rk} \mathrm{MW}(X/B) \leq 18$;
(ii) if $\operatorname{rk} \mathrm{Pic}(B) = 1$ and $B = \mathbb{P}^2$, then $\operatorname{rk} \mathrm{MW}(X/B) \leq 28$.

Proof. If $\operatorname{rk} \mathrm{Pic}(B) = 1$, B is a log del Pezzo surface, if it is smooth, then $B \simeq \mathbb{P}^2$. If $\operatorname{rk} \mathrm{Pic}(B) > 1$, B is birational to a Mori fiber space with smooth general fiber as klt surface singularities are isolated quotient singularities. The statements follow as in Theorem 6.1. □

Corollary 6.4. *Let $X \to B$ be an elliptic threefold, as in Theorem 2.9, with $\Lambda \neq 0$, $\kappa(X) = 0$ and either $\omega_X \simeq \mathcal{O}_X$ or $h^1(X, \mathcal{O}_X) \geq 1$. Then the Mordell-Weil group of X satisfies:*
(i) $\operatorname{rk} \mathrm{MW}(X/B) \leq 18$ if $B \neq \mathbb{P}^2$;
(ii) $\operatorname{rk} \mathrm{MW}(X/B) \leq 28$ if $B = \mathbb{P}^2$.

Proof. For X a threefold as in Theorem 2.9 (6), B is smooth. □

Example 6.5. Let $X = (E \times T)/G$, where E is an elliptic curve, T an elliptic K3 surface with section and G a finite group acting faithfully on E as a subgroup of translations and the action on T leaves the elliptic fibration with section and whole $H^{2,0}$ space invariant. Then $\omega_X \simeq \mathcal{O}_X$, $h^1(X, \mathcal{O}_X) \geq 1$ and $\Lambda \neq 0$. Such G and T exist, see for example [42], and $\operatorname{rk} \mathrm{MW}(X/B) \leq 18$.

This type of threefolds (without the assumption of T being elliptically fibered) are, together with Calabi-Yau, one of the 9 classes of threefolds with $\kappa(X) = 0$ [43, 44].

Remark 6.6. The threefolds X as in Example 6.5 do not satisfy the hypothesis of [8, 4] and in general of [7], which imply (birational) boundedness of the family, and hence boundedness of $\mathrm{rkMW}(X/B)$. In fact, X is not a Calabi-Yau as defined in [4], X is instead a threefold of product type [8], possibly singular and B is not rationally connected [7].

These threefolds illustrate the properties described in the Remark following Theorem 1.4 of [7]. In fact, the Mori fibration $B \to E/G$ is the maximal rationally connected (MRC, as in Definition 2.14) fibration. The induced fibration $X \to E/G$ is isotrivial. The non-rationally-connected part of the base contributes only isotrivially to the geometry, in agreement with [7]. At the same time, the presence of the elliptic curve E leads to families which are not birationally bounded, even though the Mordell–Weil rank depends only on the fibration of the isotrivial surface and remains uniformly bounded by Theorem 6.13.

Let $X \to B$ be as in Theorem 2.9, with $\kappa(X) = 0$ and B a singular log del Pezzo surface (that is B is a singular $\mathbb{Q}$-factorial surface with at worst klt singularities, $-K_B$ ample) with $\operatorname{rk}\operatorname{Pic}(B) = 1$. This is the case excluded by the hypothesis of Theorem 6.3.

Such singular log del Pezzo surfaces are classified in [45] and [46], following [47] (see also [48]). The classification provides a list of 24 non-Gorenstein families, 29 Gorenstein surfaces, defined up to isomorphisms, and an infinite family of Gorenstein surfaces [45] (see also [49] for boundedness results).

General results of [50, Thm. 6.5] and [47, Lemma 13.7] prove that B^{sm} is strongly rationally connected, that is, there exists a family of very free rational curves $\{C_t\} \subset B^{sm}$. However, these curves cannot be used to obtain bounds on $\operatorname{rk}\operatorname{MW}(X/B)$, as the following holds:

Lemma 6.7 (Lemma 13.7, [47])**.** *Let B be as above. Suppose $\mathbb{P}^1 \simeq C \subset B^{sm}$. Then either $B = \mathbb{P}^2$ or $B = \mathbb{P}[1,1,n]$ is the cone over a rational normal curve of degree n with zero section σ_0 and $C \in |\sigma_0|$.*

If $B = \mathbb{P}[1,1,n]$ and C are as in Lemma 6.7 the numerical invariants $(g(C), -K_B \cdot C)$ grow with n and $-10K_B \cdot C + 2g(C) - 2$ cannot provide a uniform bound for $\operatorname{rk}\operatorname{MW}(X/B)$.

Nevertheless, in Theorem 6.13 we will prove the existence of an explicit uniform bound. Unlike the Calabi–Yau case, where the universal bound is 28, the broader class of elliptic threefolds with $\kappa(X) = 0$ requires a larger uniform bound.

We need the following:

Lemma 6.8 (The Bogomolov Bound, Lemma 1.8.1, [47])**.** *Let B be a singular log del Pezzo surface with $\operatorname{rk}\operatorname{Pic}(B) = 1$. Then B has at most 4 singular points.*

Lemma 6.9 (Main Theorem and Corollary 4.7, [20])**.** *Let $X \to B$, as in Theorem 2.9, with $\kappa(X) = 0$ and B a singular log del Pezzo surface with $\operatorname{rk}\operatorname{Pic}(B) = 1$. Let $f : \widetilde{B} \to B$ be the minimal resolution of B. Then*

(a) $f^{-1}(P)$ *is a Hirzebruch–Jung string of type* $< n_P, q_P >$, $\forall P \in Sing(B)$,
(b) $n_P \leq 6$, $\forall P \in Sing(B)$,
(c) $2\chi(\mathcal{O}_X) = \sum(1 - 1/n_P)$.

Lemma 6.9 implies that the infinite series of singular Del Pezzo surfaces $\mathbb{P}[1, 1, n]$ of Lemma 6.7 do not occur.

Corollary 6.10. *Let* $X \to B$, *as in Theorem 2.9, with* $\kappa(X) = 0$ *and* B *a singular log del Pezzo surface with* $\operatorname{rk}\operatorname{Pic}(B) = 1$. *Then the possible sets of integers* n_p *which can occur in the basket of singularities of* B *are* $(2, 3, 6), (2, 4, 4), (3, 3, 3)$.

Moreover $\chi(\mathcal{O}_X) = 1$.

Proof. Since $0 \leq \chi(\mathcal{O}_X) \leq 4$ [Theorem 3.1, [43]], the statement follows from Lemma 6.8 and Lemma 6.9. Note that the configuration $(2, 2, 2, 2)$ is excluded by classical results originating with the work of Du Val (see for example [45]). □

Corollary 6.11. *Let* B *as in Corollary 6.10.*

(i) If B *is Gorenstein, the possible singularities of* B *are:*
$(A_5 + A_2 + A_1)$, $(A_1 + 2A_3)$, $3A_2$.
Moreover, each basket of singularities determines uniquely a weak del Pezzo surface B.

(ii) If B *is not Gorenstein, the possible baskets of singularities are:*
$(A_1, (-3), (-6))$, $(A_1, A_2, (-6))$, $(A_1, A_5, (-3))$, $(A_1, (-4), (-4))$, $(A_1, A_3, (-4))$, $((-3), (-3), (-3))$, $(A_2, A_2, (-3))$, $(A_2, (-3), (-3))$.
Moreover, there exists a finite number of families with the allowed baskets of singularities.

The singularities are cyclic quotients.

Proof. By Corollary 6.10 and [45], [47] and [46]. □

Lemma 6.12. [51], [52]

Let B *as in Corollary 6.11. There exists an integer* m *such that* $|-mK_B|$ *is base point free. In particular:*

(i) If B *is Gorenstein and it has basket* $(A_5 + A_2 + A_1)$, *then* $|-2K_B|$ *is base point free, in the other cases* $|-K_B|$ *is base point free.*

(ii) If B *is not Gorenstein then* $|-2i(B)\ K_B|$ *is base point free, where* $i(B)$ *is the Cartier index of* B.

Proof. The existence of an integer m such that $|-mK_B|$ is base point free follows from classical results of the Minimal Model Program (Kawamata-Shokurov). If B is Gorenstein Noether's formula gives $K_B^2 = 9 - e$, where e is the number of curves on the minimal resolution of B, so in case (i) $K_B^2 = 1$ and $K_B^2 \geq 2$ otherwise. (i) then follows from [51, Theorem 1, IV] and (ii) from [52, Corollary 1.5]. □

Properties of log del Pezzo surfaces with $i(B) = 2$ have been studied and classified in [53], the ones with $i(B) = 3$ in [54] and [55], also in the context of mirror symmetry and mutations.

We are now ready to state the general result:

Theorem 6.13. *Let $X \to B$ be an elliptic threefold as in Theorem 2.9, with $\kappa(X) = 0$ and $\Lambda \neq 0$. Then*

$$\text{rk MW}(X/B) \leq 810,$$

the bound is independent of X, B, and the torsion index of K_X. In addition one of the following holds:

(1) If $\rho(B) > 1$, then rk $\text{MW}(X/B) \leq 18$.
(2) If $\rho(B) = 1$ and B is smooth, then rk $\text{MW}(X/B) \leq 28$, *and $B \simeq \mathbb{P}^2$.*
(3) If $\rho(B) = 1$ and B is singular, the bounds on rk $\text{MW}(X/B)$ *are listed in Tables 1, 2 and 3.*

Proof. (1) and (2) are proved in Theorem 6.3. Let now B be as in (3), $|-mK_B|$ as in Lemma 6.12, and $C \in |-mK_B|$ be a smooth general curve. $C \subset B^{sm}$ satisfies the hypothesis of Theorem 4.15, by Bertini's Theorem, since the singularities of B are finite. Then again by Theorem 4.15 rk $\text{MW}(X/B) \leq 10C \cdot (-K_B) + 2g(C) - 2$, and this bound depends only on the basket of singularities, by Corollary 6.11 and Lemma 6.12.

TABLE 1. Numerical invariants for a general smooth curve $C \in |-K_B|$, B Gorenstein.

Basket	K_B^2	$-K_B \cdot C$	$g(C)$	$-10K_B \cdot C + 2g(C) - 2$
$(A_1 + 2A_3)$	2	2	1	20
$(3A_2)$	3	3	1	30

TABLE 2. Numerical invariants for a general smooth curve $C \in |-2K_B|$, B Gorenstein.

Basket	K_B^2	$-K_B \cdot C$	$g(C)$	$-10K_B \cdot C + 2g(C) - 2$
$(A_1 + A_2 + A_5)$	1	2	2	22

□

TABLE 3. Numerical invariants associated with a general smooth curve $C \in |-2i(B)K_B|$, B non Gorenstein.

Basket	$i(B)$	K_B^2	$-K_B \cdot C$	$g(C)$	$-10K_B \cdot C + 2g(C) - 2$
$(A_1, -3, -6)$	3	9	54	136	810
$(A_1, A_2, -6)$	3	$\frac{23}{3}$	46	116	690
$(A_1, -3, A_5)$	3	$\frac{7}{3}$	14	36	210
$(A_1, -4, -4)$	2	8	32	49	416
$(A_1, A_3, -4)$	2	5	20	31	260
$(-3, -3, -3)$	3	7	42	106	630
$(A_2, A_2, -3)$	3	$\frac{13}{3}$	26	66	390
$(A_2, -3, -3)$	3	$\frac{17}{3}$	34	86	510

6.3. $\mathrm{MW}(X/B)_{\mathrm{tors}}$ and further bounds on rkMW(X/B).

The 19 possible torsion subgroups of an elliptic fibration over a rational curve are listed in Table 4, following the results in [56].

TABLE 4.

s as in $(1, s)$ (cyclic)	$(r, s),\ r > 1$ (product)
$0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12$	$(2,2), (2,4), (2,6), (2,8), (3,3), (3,6), (4,4), (5,5)$

These bound the possibilities for the elliptic 3-folds we are interested in as well:

Theorem 6.14. *Let $X \to B$ be an elliptic 3-fold, as in Theorem 2.9, with $\kappa(X) = 0$ and $\Lambda \neq 0$.*

(1) If either $\rho(B) > 1$ or $B = \mathbb{P}^2$, the possible subgroups of $\mathrm{MW}(X/B)_{\mathrm{tors}}$ are the torsion subgroups $G_f \cong \mathbb{Z}/r \times \mathbb{Z}/s$ listed in Table 5.

(2) If $\rho(B) > 1$ and $\mathrm{MW}(X/B)_{\mathrm{tors}} \neq 0$, $\operatorname{rk} \mathrm{MW}(X/B)$ satisfies the additional bounds listed in Table 6.

TABLE 5.

	s as in $(1, s)$ (cyclic)	$(r, s),\ r > 1$ (product)
$B,\ \rho(B) > 1$	$0, 2, 3, 4, 5, 6, 7, 8$	$(2,2), (2,4), (2,6), (3,3), (4,4)$
$B = \mathbb{P}^2$	$0, 2, 3, 4, 5, 6, 9, 10$	$(2,2), (2,4), (3,3), (3,6)$

Proof. (1) For any such B there exists a smooth movable rational curve $C \subset B^{sm}$, [50]. We can choose a general C such that $Z = \pi^{-1}(C)$ is a smooth (minimal) surface and $\mathrm{MW}(X/B)_{\mathrm{tors}} \subseteq \mathrm{MW}(Z/C)_{\mathrm{tors}}$,

TABLE 6.

	s as in $(1,s)$ (cyclic)							$(r,s),\ r>1$ (product)				
G_f	2	3	4	5	6	7	8	(2,2)	(2,4)	(2,6)	(3,3)	(4,4)
$\max \operatorname{rk} \mathrm{MW}$	10	6	4	2	4	0	0	6	4	0	2	0

by Proposition 2.18. The table is then derived from [56, Lemma1.1, Table 4.5]; note that (r,s) is determined by $K_B \cdot C$.

(2) In [57], the possible values of $\operatorname{rk} \mathrm{MW}(X/B)$ are determined for the case in which $Z = \pi^{-1}(C)$ is a smooth K3 surface, which is the case whenever $\rho(B) > 1$.

□

6.4. $\kappa(X) < 0$.

Similarly, we can obtain explicit bounds on the Mordell-Weil rank of elliptic threefolds X with $\kappa(X) < 0$:

Proposition 6.15. *Let $X \to B$ be an elliptic threefold as in Theorem 2.9, with $\kappa(X) < 0$ and $\Lambda \neq 0$. Then:*

(1) If $\rho(B) > 1$, $\operatorname{rk} \mathrm{MW}(X/B) \leq 8$.

(2) If $\rho(B) = 1$ and B is smooth, $\operatorname{rk} \mathrm{MW}(X/B) \leq 18$.

Proof. Since $\kappa(X) < 0$, $K_B + \Lambda < 0$. In case (1) the general curve C of the Mori fibration is rational with $\Lambda \cdot C = 1$, and $Z \to C$ is a rational elliptic surface. In case (2), $B \simeq \mathbb{P}^2$, either $\Lambda \cdot C = 1$ or 2 and $Z \to C$ is either a rational or K3 elliptic surface. □

Example 6.16. Relevant examples of this type of threefolds are in the papers [58] and [59].

7. Bounds for the Mordell-Weil rank of elliptic fourfolds with $\kappa(X) \leq 0$

In this section, we prove bounds for elliptic fourfolds with $K_X \equiv 0$.

We work under the following hypothesis:

Assumption 7.1. *Let $X \to B$ as in Theorem 2.9, with $K_X \equiv 0$ such that $\Lambda = -K_B \neq 0$. Let $B \dashrightarrow B' \to U$ as in Definition 2.11.*

(a) If $\dim U = 1$, we assume that B' has terminal singularities.

(b) If $\dim U = 0$, that is $\rho(B') = 1$, we assume that B' is smooth and furthermore that it is general in its deformation family if in addition $r(B') = 1$, where $r(B')$ is the Fano index of K_B, i.e. $-K_{B'} = r(B')H$ with H the generator of $\operatorname{Pic}(B')$.

Remark 7.2. (a) The assumption that B' has terminal singularities is a natural generalization of the lower dimensional case, where B is smooth.

(b) If B' is Fano with Gorenstein terminal singularities, then B' is smoothable [60], so if B' is Gorenstein and general, the assumption is satisfied.

(c) There are some results on the existence of free curves on Fano klt varieties [61].
(d) There exists a partial classification of factorial Fano threefolds B' with terminal singularities in [62], see also [63] and [64]. Recall that $\mathbb{Q}$-factorial Gorenstein is factorial.

We prove the following:

Theorem 7.3. *Under the hypothesis as in Assumption 7.1,*

$$\operatorname{rk}\operatorname{MW}(X/B) \leq 38.$$

In addition, if $B' \to U$, is a Mori fiber space with $\dim U > 0$,

(i) $\operatorname{rk}\operatorname{MW}(X/B) \leq 28$ *when the general fiber of the Mori fibration is $\mathbb{P}^2$,*
(ii) and otherwise, $\operatorname{rk}\operatorname{MW}(X/B) \leq 18$.

The proof proceeds by considering separately the three possible dimensions of U of the Mori fiber space $B' \to U$. In each case we construct a curve C satisfying the hypotheses of Theorem 4.17.

Corollary 7.4. *In particular, the bounds of Theorem 7.3 apply to elliptic Calabi-Yau fourfolds with the additional Assumption 7.1.*

Remark 7.5. In addition to Calabi-Yau fourfolds in the sense of Definition 2.1, the bounds of Theorem 7.3 apply also to other fourfolds, for example fourfolds of the form $(E \times T)/G$, where one of the following holds (such examples exist, as in Example 6.5):

(1) E is an elliptic curve, T an elliptic Calabi-Yau threefold with section and G a finite group acting faithfully on E as a subgroup of translations and the action on T preserves the elliptic fibration with section and leaves $H^{3,0}$ invariant;
(2) T is an abelian surface and E is an elliptic K3 surface with section and G a finite group acting faithfully on T as a subgroup of translations and the action on E leaves invariant the elliptic fibration with section and $H^{2,0}$.

Remark 7.6. The fourfolds described in Remark 7.5 have the same isotrivial behaviour as Example 6.5 and Remark 6.6, and in [7]. In case (1) $X = (E \times T)/G$, where $T \to B_0$ is an elliptic Calabi–Yau threefold. The base of the induced elliptic fibration $B = (E/G) \times B_0$ is uniruled but not rationally connected. The MRC fibration is given by the projection $B \dashrightarrow E/G$, whenever B_0 is uniruled, and the induced map $X \dashrightarrow E/G$ has fibers isomorphic to T. Hence this fibration is isotrivial.

A similar description applies to case (2), where the abelian surface factor plays the role of the non-rationally-connected base.

In both cases, the non-rationally-connected directions contribute only isotrivially, while the variation relevant for the Mordell–Weil group is governed by the geometry of the elliptic fibration on the remaining factor.

This explains why, while these families are not birationally bounded, the Mordell–Weil rank remains uniformly bounded as in Theorem 7.3.

The same reasoning, with Assumption 7.1, provides bounds on the Mordell-Weil rank of elliptic fourfolds with $\kappa(X) < 0$, as in Example 6.16.

We reduce the proof of Theorem 7.3, to the existence of suitable curves $C \subset B$ satisfying the hypotheses of either Theorems 3.1, 4.15 or 4.17:

Theorem 7.7. *Let B' as in Theorem 2.12, with B' smooth. Then there exists a smooth curve $C \subset B$ such that* $\operatorname{rk}\mathrm{MW}(X/B) \leq 10(-K_{B'} \cdot C) + 2g(C) - 2$.

Proof. Theorem 2.12 implies that in cases (1) and (2) there exists a curve C smooth, movable and rational such that Z is smooth, and Theorems 3.1 and 4.15 both apply. Otherwise, in case (3), $\operatorname{rk}\mathrm{Pic}(B') = 1$ and we show that there exists a smooth curve C such that either $Z = \pi^{-1}(C)$ is smooth or Z is birationally a K3 surface and Theorem 4.17 applies.

We prove the existence of such a curve C in the sections that follow. □

Corollary 7.8. *Let B, B' and $\mu : B \to B'$ be as in Assumption 7.1. Then there exists a smooth curve C in B such that*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 10(-K_B \cdot C) + 2g(C) - 2.$$

Proof. By the structure Theorem 2.12 if $\rho(B) > 1$, either cases (1) and (2) apply and the curve C can be taken in the open set where μ is an isomorphism. If $\rho(B) = 1$, then $B = B'$. □

We then prove Theorem 7.3 by combining the optimal bounds obtained via Theorem 7.7 for all possible B under the stated assumptions.

Proposition 7.9. *In case (1) of Theorem 2.12*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 18.$$

Proof. Since B' has at worst klt singularities, the general fiber of $f : B' \to U$ is in fact a smooth rational movable curve with Z smooth and if $\Lambda \neq 0$, Theorem 4.15 and adjunction imply: $\operatorname{rk}\mathrm{MW}(X/B) \leq -10(C \cdot K_B) - 2 = 18$. □

Proposition 7.10. *In case (2) of Theorem 2.12,* $\operatorname{rk}\mathrm{MW}(X/B) \leq 28$ *if the fiber is $\mathbb{P}^2$ and* $\operatorname{rk}\mathrm{MW}(X/B) \leq 18$ *otherwise.*

Proof. By Assumption 7.1, the general fiber F of $f : B' \to U$ is a smooth del Pezzo surface, as terminal singularities are isolated, and $\pi^{-1}(F)$ is a smooth threefold with Kodaira dimension zero and conclude as in Theorem 6.3. □

Theorem 7.11. *In case (1) and (2) of Theorem 2.12, the torsion groups* $\mathrm{MW}(X/B)_{\mathrm{tors}}$ *are those listed in the tables of Theorem 6.14.*

Proof. We argue as in Propositions 7.9 and 7.10 and conclude as in Theorem 6.14. □

In case (3) of Theorem 2.12 B is a Fano threefold, which we assume to be smooth, as in Assumption 7.1 with $\rho(B) = 1$. As in the proof of Corollary 7.8, we can set $B = B'$.

Let $r(B)$ be the index of K_B, i.e. $-K_B = r(B)H$ with H the generator of $\mathrm{Pic}(B)$. Recall the following, by now classic, result:

Theorem 7.12. [65, Table 6.5] *Let B be a smooth Fano threefold with $\rho(B) = 1$ and index $r(B)$.*

(i) If $r(B) = 3, 4$, then $B = \mathbb{P}^3$ or a quadric $Q \subset \mathbb{P}^4$.

(ii) If $r(B) = 2$, then there are 5 deformation types and B is called a del Pezzo threefold.

(iii) If $r(B) = 1$, then $H^3 = -K_B^3$ and there are 10 deformation types. Furthermore $4 \leq H^3 \leq 18$ unless:

- *$H^3 = 2$ and B is a double cover of $\mathbb{P}^3$ with branch locus a divisor of degree 6,*
- *$H^3 = 22$ and B is the zero locus of $(\wedge^2 \mathcal{U}^v)^{\oplus 3}$ on $Gr(3, 7)$.*

In addition $-K_B$ is very ample unless

(a) $H^3 = 2$ and B is a double cover of $\mathbb{P}^3$ with branch locus a divisor of degree 6,

(b) $H^3 = 4$ and B is the double cover of a quadric in $\mathbb{P}^4$ branched along an octic,

(c) $H^3 = 10$ and B is the double cover of a section of Plücker embedding of $Gr(2, 5)$.

We now analyze the cases in turn.

7.1. Case (i): $\rho(B) = 1$, $r(B) = 3, 4$.

Proposition 7.13. *Let B be a smooth Fano threefold with $\rho(B) = 1$, $r(B) = 3, 4$. Then*

$$\mathrm{rk}\,\mathrm{MW}(X/B) \leq 38.$$

Proof. Each such Fano contains a movable line ℓ, that is a smooth rational curve with $H \cdot \ell = 1$, for H the ample generator of $\mathrm{Pic}(B)$, such that $Z = \pi^{-1}(\ell)$ is smooth. Then $-K_B \cdot \ell = r(B)$, and

$$\mathrm{rk}\,\mathrm{MW}(X/B) \leq 10(-K_B \cdot \ell) - 2 \leq 38.$$

□

7.2. Case (ii): $\mathrm{rk}\,\mathrm{Pic}(B) = 1$, $r(B) = 2$.

In the following Propositions 7.14 and 7.15 we recall properties of a del Pezzo threefold B (that is B Fano, such that $r(B) = 2$ and $\rho(B) = 1$) up to deformation type:

Proposition 7.14. [66, Proposition 3.21] *Let B be a smooth del Pezzo threefold such that $(-K_B)^3 \geq 32$. Then*

(1) there exists a line $\ell \subset B$ through any point in B,

(2) $F_1(B)$, *the Hilbert scheme parametrizing lines on* B, *is reduced, non-singular, and each of its components is two-dimensional,*
(3) *a line* ℓ *corresponding to the general point of a component of* $F_1(B)$ *has normal bundle* $N_{\ell/B} = \mathcal{O}_\ell \oplus \mathcal{O}_\ell$.

Proposition 7.15. [66, Table 1] *Let* B *be a smooth del Pezzo threefold such that* $(-K_B)^3 < 32$. *Then*

(i) *If* $(-K_B)^3 = 24$, $B \subset \mathbb{P}^4$ *is a cubic,*
(ii) *if* $(-K_B)^3 = 16$ *and* B *is a double cover of* $\mathbb{P}^3$ *branched along a smooth quartic,*
(iii) *if* $(-K_B)^3 = 8$ *and* B *is the double cover of the cone over the Veronese surface (or equivalently a hypersurface of degree* 6 *in* $\mathbb{P}(1,1,1,2,3)$).

Corollary 7.16. *Let* B *be a del Pezzo threefold, then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 38.$$

If $\Sigma \subset B$ *is general (in the relevant linear system),* $\operatorname{rk}\mathrm{MW}(X/B) \leq 18$.

Proof. If $(-K_B)^3 \geq 32$ by Proposition 7.14 there exists a line ℓ with $-K_B \cdot \ell = 2$, $\ell \not\subset \Sigma$. Hence Z is (birationally) a K3 and $\operatorname{rk}\mathrm{MW}(X/B) \leq 18$, by Theorem 4.17.

Assume now that $(-K_B)^3 < 32$, and let us analyze the three cases of Proposition 7.15.

In case (i), $B \subset \mathbb{P}^4$ is any smooth cubic, the Fano variety of lines is two dimensional, the general line has normal bundle $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}$; by direct computation for every point $p \in B$ there are six lines. Then there exists a line $\ell \not\subset \Sigma$ with $-K_B \cdot \ell = 2$. Hence Z is (birationally) a K3 and $\operatorname{rk}\mathrm{MW}(X/B) \leq 18$, by Theorem 4.17.

In case (ii), the double cover of a conic with 4 (distinct) bitangent points is a conic C (at the ramification points the local equation is $y^2 = x^2$). The family of such conics dominates B, it is base point free, the general conic is smooth (the general such conic C has either normal bundle $\mathcal{O}_X(1) \oplus \mathcal{O}_X(1)$ or $\mathcal{O}_X \oplus \mathcal{O}_X(2)$ [67, Lemma 2.2]). We then get the bound: $\operatorname{rk}\mathrm{MW}(X/B) \leq 40 - 2 = 38$. The lines in B are the inverse image of bitangent lines to the quartic surface in $\mathbb{P}^3$. The lines do not cover B. If $C \subset B$ is a general line, for *general* (B, Σ), $\operatorname{rk}\mathrm{MW}(X/B) \leq 18$ by Proposition 4.11 (and $N_{C/X} = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}$ see also [67, Lemma 2.1]).

In case (iii), the conics are dominant very free movable curves [68, Proposition 3.2]. Then, $\operatorname{rk}\mathrm{MW}(X/B) \leq 40 - 2 = 38$. □

7.3. B smooth Fano threefold with $\operatorname{rk}\mathrm{Pic}(B) = 1$, $r(B) = 1$.

7.3.1. $-K_B$ *is very ample.*

We start with the following:

Proposition 7.17. *Let* $-K_B$ *be very ample with* $4 \leq (-K_B)^3 \leq 18$. *Assume in addition that if* $4 \leq (-K_B)^3 \leq 8$ *then* B *is general in moduli. Then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 38.$$

Proof. The results in [69, 70, Theorem 1 and Lemma 8.1] imply that there exists a dominant family of smooth very free rational curves C of degree 4 on B; it follows that there exists a smooth surface Z with $Z \to C$ and $\operatorname{rk}\mathrm{MW}(X/B) \leq \operatorname{rk}\mathrm{MW}(Z/C) \leq 40 - 2$. □

Next we consider the cases of $(-K_B)^3 = 22$ and the three cases when $-K_B$ is not very ample.

Proposition 7.18. *If $(-K_B)^3 = 22$, then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 10(-K_B \cdot C) - 2 \leq 18.$$

Proof. There exists a 1-dimensional family of lines and a 2-dimensional family of smooth conics [71, Lemma 3.2]. Through any point $p \in B$ there exists at least a conic C. Since $-K_B \cdot C = 2$, Proposition 4.11 and Theorem 4.17 apply. □

7.3.2. *$-K_B$ is not very ample.*

Then B is one of the cases treated in the following three propositions:

Proposition 7.19. *Let B be a hypersurface of degree 6 in $\mathbb{P}(1^4, 3)$ and $(-K_B)^3 = 2$, equivalently B is the double cover of $\mathbb{P}^3$ branched over a surface of degree 6. Then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 20 + 2 = 22\,.$$

Proof. The double cover of the family of lines is a movable family of curves of genus 2, with general smooth member C. Then Z can be chosen to be smooth and $\operatorname{rk}\mathrm{MW}(X/B) \leq 10(-K_B \cdot C) + 2g(C) - 2 = 20 + 2 \leq 22$. □

Proposition 7.20. *Let B be the double cover of a general quadric in $\mathbb{P}^4$ branched along an octic. Then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 20.$$

Proof. The double covers of the family of lines, is a movable base point free family of curves of genus 1, with general smooth member C. Then Z can be chosen to be smooth and

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 10(-K_B \cdot C) + 2 - 2 \leq 20\,.$$

□

Proposition 7.21. *B is a special Gushel-Mukai threefold (the general double cover of a section of a Plücker embedding of $Gr(2, 5)$ by a codimension 3 subspace) and $(-K_B)^3 = 10$. Then*

$$\operatorname{rk}\mathrm{MW}(X/B) \leq 18.$$

Proof. By construction (B is the double cover of a $\mathbb{P}^2$-bundle) there exists a family of conics which dominates B; the family has dimension two [72, Theorem 7.3, Remark 7.4]. We conclude by Proposition 4.11 and Theorem 4.17. □

8. Bounds for elliptic varieties with $\omega_X \simeq \mathcal{O}_X$, of any dimension

Theorems 4.15, 6.1 and Theorem 7.3 with the Assumption 7.1 can be stated as

Theorem 8.1. *Let $X \to B$ be as in Theorem 2.12, with $\omega_X \simeq \mathcal{O}_X$, $\dim X = 3, 4$. Assume that Assumption 7.1 is satisfied if $\dim X = 4$. Then*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10 \cdot (\dim B + 1) - 2 \leq 10 \cdot \dim X - 2.$$

In addition, if B is birationally a Mori fiber space $B' \to U$ with $\dim U > 0$ and general fiber F,

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10 \cdot r(F) - 2 \leq 10 \cdot (\dim F + 1) - 2 \leq 10 \cdot \dim X - 2.$$

Here the Fano index $r(F)$ is the largest integer a such that $-K_F = aH$ for H an ample divisor, since F is smooth, in particular Gorenstein. Also, for a smooth (Gorenstein) Fano variety F, the highest index is $r(F) = \dim F + 1$ and it is attained for the projective space [73].

We also have

Theorem 8.2. *Let $X \to B$ be as in Theorem 2.12, $\Lambda \neq 0$ and X Calabi-Yau. Assume there exists a smooth rational curve $C \subset B^{sm}$ and one of the following holds:*

(a) there is an (irreducible) nonsingular complex variety T, together with a smooth morphism $\varphi : T \times \mathbb{P}^1_{\mathbb{C}} \to B$ with dense image, contained in the smooth locus of B. Regarding this as a family of morphisms $\varphi_t : \mathbb{P}^1_{\mathbb{C}} \to B$, assume further that these are all embeddings, giving a family $C_t \hookrightarrow B$ of smooth rational curves, which has no base points; let $C = C_t$ be a general curve in the family;
(b) C is movable;
(c) $\rho(B) = 1$;

and that in cases (b) and (c) either $Z = \pi^{-1}(C)$ is smooth or (birationally) a K3 surface.

Then

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10(-C \cdot K_B) - 2.$$

Proof. The statement follows from Theorems 3.1, 4.15 and 4.17. □

Remark 8.3. If (B, Λ) has log canonical singularities, and $-K_B \cdot C > \dim B$ for all rational curve $C \subset B$, then $B \simeq \mathbb{P}^{\dim B}$ ([74], which generalizes [75]). Then there always exists a rational curve $C \subset B$, B as in Theorem 2.9, with $(-C \cdot K_B) \leq (\dim B + 1)$. If such a curve satisfies one of the hypotheses of Theorem 8.2, then

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10(-C \cdot K_B) - 2 \leq 10 \cdot (\dim B + 1) - 2 = 10 \cdot (\dim X) - 2. \tag{13}$$

However, it is not known that curves with the properties in Theorem 8.2 and $(-C \cdot K_B) \leq (\dim B + 1)$ exist when $\dim X > 3$. On the other hand, as

in the proofs of Theorem 7.3, it is also natural to consider curves of higher genus which, as we have proved, also provide the expected bounds.

By combining Theorems 8.1 and 8.2 with equation 13 in Remark 8.3 and extrapolating from our results, it is natural to make the following

Conjecture 8.4. *Let $X \to B$ be as in Theorem 2.12, X Calabi-Yau. Then B is birationally a Mori fiber space $B' \to U$ with fiber F and*

$$\operatorname{rk} \mathrm{MW}(X/B) \leq 10 \cdot (\dim F + 1) - 2 \leq 10 \cdot \dim X - 2.$$

Recall that B is birationally a Mori fiber space if $\dim X \leq 5$, or if the standard conjectures of the minimal model program hold (Theorem 2.9).

Appendix A. Bounds from physics

To every elliptic Calabi-Yau 3-fold X one can associate a six-dimensional $N = 1$ supergravity theory obtained as the low-energy limit of F-theory [76, 77, 78] compactified on X. In this context, the free part of the Mordell-Weil group of X encodes the continuous abelian gauge symmetries in the effective action, while its torsion part constrains the first fundamental group of the non-abelian gauge group (see, for example, the reviews [79, 80] for details and references). Bounds on the rank of the Mordell-Weil group, or in physics terms on the rank $r_{\mathrm{U}(1)}$ of the abelian gauge group, are motivated by general arguments suggesting that the number of light degrees of freedom and their interactions in any consistent quantum gravitational theory are severely restricted. While gauge and gravitational anomalies alone do admit infinite families of theories with arbitrarily high abelian gauge rank [81], taking into consideration constraints from quantum consistency of probe strings [82] sets tight bounds on the ranks [12, 13, 14, 83].

In [12] bounds on $r_{\mathrm{U}(1)}$ have been obtained for general $N = 1$ supergravity theories in six dimensions by analysing probe consistency of certain solitonic strings coupling to gravity. These bounds are given by $r_{\mathrm{U}(1)} \leq 22$ for theories with $T \geq 1$ tensor multiplets (subsequently improved to $r_{\mathrm{U}(1)} \leq 20$ in [83]) and $r_{\mathrm{U}(1)} \leq 32$ for $T = 0$. In explicit realisations of the supergravity via F-theory, the solitonic strings arise from D3-branes wrapping suitable curves on B. If the curve C is rational, movable and without base points on B, the associated string couples to all abelian gauge groups because C has a non-zero intersection number with the height pairing divisor associated with the Shioda map. This follows by extending arguments in [30] to smooth Calabi-Yau threefolds [15] and led, for smooth base B, to the conservative bounds $\operatorname{rk}(\mathrm{MW}(X/B)) \leq 32$ for $B = \mathbb{P}^2$ and $\operatorname{rk}(\mathrm{MW}(X/B)) \leq 20$ otherwise in [12]. In [13] these bounds were sharpened to

$$\operatorname{rk}(\mathrm{MW}(X/B)) \leq 10\, C \cdot (-K_B) - 2\,, \tag{14}$$

which also holds for elliptic Calabi-Yau fourfolds with a smooth base B and C rational, smooth and in a movable class without base points. For elliptic threefolds, (14) implies the stronger bound $\operatorname{rk}(\mathrm{MW}(X/B)) \leq 28$ if $B = \mathbb{P}^2$ (leading to $T = 0$) and $\operatorname{rk}(\mathrm{MW}(X/B)) \leq 18$ otherwise. The physics

derivation of (14) relies on a specific interpretation of the zero modes on the probe strings. For X an elliptic threefold with rationally fibered base B, [14] claims the bound $\mathrm{rk}(\mathrm{MW}(X/B)) \leq 18$ and, by combining with [40], a list of the possible torsion groups.

References

[1] D. A. Cox, *Mordell-weil groups of elliptic curves over $C(t)$ with $p_g = 0$ or 1*, *Duke Mathematical Journal* **49** (1982) 677.

[2] M. Kuwata, *Elliptic K3 surfaces with given Mordell-Weil rank*, *Comment. Math. Univ. St. Paul.* **49** (2000) 91.

[3] R. Kloosterman, *Elliptic K3 surfaces with geometric Mordell–Weil rank 15*, *Canad. Math. Bull.* **50** (2007) 215 [math/0502439].

[4] S. Filipazzi, C. D. Hacon and R. Svaldi, *Boundedness of elliptic Calabi-Yau threefolds*, *J. Eur. Math. Soc.* **27** (2025) 3583 [2112.01352].

[5] M. Gross, *A finiteness theorem for elliptic Calabi-Yau threefolds*, *Duke Math. J.* **74** (1994) 271 [alg-geom/9305002].

[6] P. Engel, S. Filipazzi, F. Greer, M. Mauri and R. Svaldi, *Boundedness of some fibered k-trivial varieties*, 2507.00973.

[7] C. Birkar, G. Di Cerbo and R. Svaldi, *Boundedness of elliptic Calabi–Yau varieties with a rational section*, *Journal of Differential Geometry* **128** (2024) [2010.09769].

[8] G. Di Cerbo and R. Svaldi, *Birational boundedness of low-dimensional elliptic Calabi-Yau varieties with a section*, *Compos. Math.* **157** (2021) 1766 [1608.02997].

[9] S. Filipazzi and F. Xu, *On the boundedness of elliptic Calabi-Yau 4-folds*, 2607.27048.

[10] A. Grassi and T. Weigand, *Elliptic threefolds with high Mordell–Weil rank*, *Commun. Num. Theor. Phys.* **16** (2022) 733 [2105.02863].

[11] N. D. Elkies, *K3 surfaces and elliptic fibrations in number theory*, *BIRS workshop Geometry and Physics of F-theory, Banff, workshop 18w5190* (2018) .

[12] S.-J. Lee and T. Weigand, *Swampland bounds on the Abelian gauge sector*, *Phys. Rev. D* **100** (2019) 026015, 10 [1905.13213].

[13] L. Martucci, N. Risso and T. Weigand, *Quantum gravity bounds on $n = 1$ effective theories in four dimensions*, *J. High Energy Phys.* (2023) Paper No. 197, 85 [2210.10797].

[14] S.-J. Lee and P.-K. Oehlmann, *Geometric bounds on the 1-form gauge sector*, *Phys. Rev. D* **108** (2023) 086021 [2212.11915].

[15] S.-J. Lee, D. Regalado and T. Weigand, *6d SCFTs and* U(1) *flavour symmetries*, *J. High Energy Phys.* (2018) 147, front matter+50 [1803.07998].

[16] S. Boucksom, J.-P. Demailly, M. Paun and T. Peternell, *The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension*, *J. Algebraic Geom.* **22** (2013) 201 [math/0405285].

[17] O. Debarre, *Rational curves on algebraic varieties*, *Lecture notes for the GAeL XVIII conference, Coimbra, and the Semaine Spéciale Master de Mathématiques, Université de Strasbourg* (2011) .

[18] K. Kodaira, *On Compact Analytic Surfaces. II, III*, *Annals of Mathematics* **77** (1963) 563.

[19] A. Grassi, *On minimal models of elliptic threefolds*, *Math. Ann.* **290** (1991) 287.

[20] A. Grassi, *The singularities of the parameter surface of a minimal elliptic threefold*, *Internat. J. Math.* **4** (1993) 203 [alg-geom/9202025].

[21] A. Grassi and D. Wen, *Higher dimensional elliptic fibrations and Zariski decompositions*, *Commun. Contemp. Math.* **24** (2022) 2150024 [1904.02779].

[22] A. Grassi, *Log contractions and equidimensional models of elliptic threefolds*, *J. Algebraic Geom.* **4** (1995) 255 [alg-geom/9305003].
[23] K. Matsuki, *Introduction to the Mori program*, Universitext. Springer-Verlag, New York, 2002, 10.1007/978-1-4757-5602-9.
[24] F. Campana, *Connexité rationnelle des variétés de Fano*, *Ann. Sci. École Norm. Sup.* **25** (1992) 539.
[25] J. Kollár, Y. Miyaoka and S. Mori, *Rationally connected varieties*, *Journal of Algebraic Geometry* **1** (1992) 429.
[26] R. Piene, *A proof of Noether's formula for the arithmetic genus of an algebraic surface*, *Compositio Math.* **38** (1979) 113.
[27] P. Deligne, *Théorème de Lefschetz et critères de dégénérescence de suites spectrales*, *Publications Mathématiques de l'IHÉS* **35** (1968) 107.
[28] T. Shioda, *On the Mordell-Weil lattices*, *Comment. Math. Univ. St. Paul.* **39** (1990) 211.
[29] D. R. Morrison and D. S. Park, *F-Theory and the Mordell-Weil Group of Elliptically-Fibered Calabi-Yau Threefolds*, *JHEP* **10** (2012) 128 [1208.2695].
[30] D. A. Cox and S. Zucker, *Intersection numbers of sections of elliptic surfaces*, *Invent. Math.* **53** (1979) 1.
[31] R. Friedman, J. Morgan and E. Witten, *Vector bundles and* F *theory*, *Comm. Math. Phys.* **187** (1997) 679 [hep-th/9701162].
[32] K. Oguiso, *Shioda-Tate formula for an abelian fibered variety and applications*, *J. Korean Math. Soc.* **46** (2009) 237 [math/0703245].
[33] R. Wazir, *Arithmetic on elliptic threefolds*, *Compositio Mathematica* **140** (2004) 567–580 [math/0112259].
[34] R. Miranda, *The moduli of Weierstrass fibrations over* $\mathbb{P}^1$, *Math. Ann.* **255** (1981) 379.
[35] W. Fulton, *Intersection Theory*, vol. 2 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*. Springer, New York, 2 ed., 1998, 10.1007/978-1-4612-1700-8.
[36] B. Conrad, *Chow's K/k-image and K/k-trace, and the Lang–Néron theorem*, *L'Enseignement Mathématique* **52** (2006) 37.
[37] S. Lang, *Abelian Varieties*, no. 7 in Interscience Tracts in Pure and Applied Mathematics. Interscience Publishers, Inc., New York, 1959.
[38] J. H. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, vol. 151 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1994, 10.1007/978-1-4612-0851-8.
[39] B. Kahn, *Sur le groupe de classes d'un schéma arithmétique*, *Bulletin de la Société Mathématique de France* **134** (2006) 395.
[40] N. Hajouji and P.-K. Oehlmann, *Modular curves and Mordell-Weil torsion in F-theory*, *JHEP* **04** (2020) 103 [1910.04095].
[41] A. R. Mavlyutov, *Semiample hypersurfaces in toric varieties*, *Duke Mathematical Journal* **101** (2000) 85 [math/9812163].
[42] A. Garbagnati and A. Sarti, *Elliptic fibrations and symplectic automorphisms on K3 surfaces*, *Communications in Algebra* **37** (2009) 3601 [0801.3992].
[43] Y. Kawamata, *On the plurigenera of minimal algebraic* 3*-folds with* $K \equiv 0$, *Math. Ann.* **275** (1986) 539.
[44] D. R. Morrison, *A remark on Kawamata's paper "On the plurigenera of minimal algebraic 3-folds with* $K \equiv 0$*"*, *Mathematische Annalen* **275** (1986) 547.
[45] Q. Ye, *On Gorenstein log del Pezzo surfaces*, *Japan. J. Math. (N.S.)* **28** (2002) 87 [math/0109223].
[46] J. Lacini, *On rank one log del Pezzo surfaces in characteristic different from two and three*, *Adv. Math.* **442** (2024) Paper No. 109568, 77 [2005.14544].
[47] S. Keel and J. McKernan, *Rational curves on quasi-projective surfaces*, *Mem. Amer. Math. Soc.* **140** (1999) viii+153 [alg-geom/9707016].

[48] M. Nagaoka, *A review of Lacini's classification of rank one log del Pezzo surfaces in characteristic different from two and three*, 2512.12294.
[49] V. Alexeev, *Boundedness and K^2 for log surfaces*, *Internat. J. Math.* **5** (1994) 779 [alg-geom/9402007].
[50] C. Xu, *Strong rational connectedness of surfaces*, *J. Reine Angew. Math.* **665** (2012) 189 [0810.2597].
[51] M. Demazure, *Surfaces de del pezzo—iv. systèmes anticanoniques*, in *Séminaire sur les Singularités des Surfaces*, M. Demazure, H. C. Pinkham and B. Teissier, eds., vol. 777 of *Lecture Notes in Mathematics*, (Berlin, Heidelberg), pp. 50–60, Springer, (1980), DOI.
[52] O. Fujino, *Effective basepoint-free theorem for semi-log canonical surfaces*, *Publ. Res. Inst. Math. Sci.* **53** (2017) 349 [1509.07268].
[53] V. A. Alekseev and V. V. Nikulin, *Classification of del Pezzo surfaces with log-terminal singularities of index ≤ 2 and involutions on $K3$ surfaces*, *Dokl. Akad. Nauk SSSR* **306** (1989) 525.
[54] K. Fujita and K. Yasutake, *Classification of log del Pezzo surfaces of index three*, *J. Math. Soc. Japan* **69** (2017) 163 [1401.1283].
[55] A. Corti and L. Heuberger, *Del Pezzo surfaces with $\frac{1}{3}(1,1)$ points*, *Manuscripta Math.* **153** (2017) 71 [1505.02092].
[56] R. Miranda and U. Persson, *Torsion groups of elliptic surfaces*, *Compositio Math.* **72** (1989) 249.
[57] I. Shimada, *On elliptic $K3$ surfaces*, *Michigan Math. J.* **47** (2000) 423 [math/0505140].
[58] B. Totaro, *Hilbert's fourteenth problem over finite fields, and a conjecture on the cone of curves*, *Compositio Mathematica* **144** (2008) 1176 [0808.0695].
[59] A. Prendergast-Smith, *The cone conjecture for some rational elliptic threefolds*, *Mathematische Zeitschrift* **272** (2012) 589 [0910.5888].
[60] Y. Namikawa, *Smoothing Fano 3-folds*, *Journal of Algebraic Geometry* **6** (1997) 307.
[61] E. Jovinelly, B. Lehmann and E. Riedl, *Free curves and fundamental groups*, 2510.27031.
[62] A. Kuznetsov and Y. Prokhorov, *On higher-dimensional del Pezzo varieties*, *Izvestiya Math.* **87** (2023) 488 [2206.01549].
[63] A. Bayer, A. Kuznetsov and E. Macrì, *Mukai models of Fano varieties*, *J. Reine Angew. Math.* (2026) [2501.16157].
[64] A. G. Kuznetsov and Y. G. Prokhorov, *1-nodal Fano threefolds with Picard number 1*, *Izv. Ross. Akad. Nauk Ser. Mat.* **89** (2025) 80 [2312.13782].
[65] V. A. Iskovskikh, *Fano threefolds. II*, *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* **42** (1978) 506.
[66] Y. G. Prokhorov, *Fano threefolds*, *Proceedings of the Steklov Institute of Mathematics* **328** (2025) S1–S130 [2511.22197].
[67] D. G. Markushevich and A. S. Tikhomirov, *A parametrization of the theta divisor of the quartic double solid*, *Int. Math. Res. Not.* (2003) 2747 [math/0212148].
[68] N. Shimizu and S. Tanimoto, *The spaces of rational curves on del Pezzo threefolds of degree one*, *Eur. J. Math.* **8** (2022) 291 [2101.04312].
[69] B. Lehmann and S. Tanimoto, *Rational curves on prime Fano threefolds of index 1*, *J. Algebraic Geom.* **30** (2021) 151 [1808.05590].
[70] E. Jovinelly, B. Lehmann and S. Tanimoto, *Corrigenda to "rational curves on prime Fano threefolds of index 1"*, *J. Algebraic Geom.* **34** (2025) 535.
[71] E. Arrondo and D. Faenzi, *Vector bundles with no intermediate cohomology on Fano threefolds of type V_{22}*, *Pacific Journal of Mathematics* **225** (2006) 201.
[72] O. Debarre and A. Kuznetsov, *Quadrics on Gushel-Mukai varieties*, 2409.03528.
[73] S. Kobayashi and T. Ochiai, *Characterizations of complex projective spaces and hyperquadrics*, *Journal of Mathematics of Kyoto University* **13** (1973) 31 .

[74] O. Fujino, E. Jovinelly, B. Lehmann and E. Riedl, *A characterization of projective space via lengths of extremal rays*, 2602.22192.
[75] K.-H. Cho, Y. Miyaoka and N. I. Shepherd-Barron, *Characterizations of projective space and applications to complex symplectic manifolds*, *Higher Dimensional Varieties and Rational Points* **12** (2003) 1.
[76] C. Vafa, *Evidence for F theory*, *Nucl. Phys. B* **469** (1996) 403 [hep-th/9602022].
[77] D. R. Morrison and C. Vafa, *Compactifications of F theory on Calabi-Yau threefolds. 1*, *Nucl. Phys. B* **473** (1996) 74 [hep-th/9602114].
[78] D. R. Morrison and C. Vafa, *Compactifications of F theory on Calabi-Yau threefolds. 2.*, *Nucl. Phys. B* **476** (1996) 437 [hep-th/9603161].
[79] T. Weigand, *F-theory*, *PoS* **TASI2017** (2018) 016 [1806.01854].
[80] M. Cvetič and L. Lin, *TASI Lectures on Abelian and Discrete Symmetries in F-theory*, *PoS* **TASI2017** (2018) 020 [1809.00012].
[81] D. S. Park and W. Taylor, *Constraints on 6D Supergravity Theories with Abelian Gauge Symmetry*, *JHEP* **01** (2012) 141 [1110.5916].
[82] H.-C. Kim, G. Shiu and C. Vafa, *Branes and the Swampland*, *Phys. Rev. D* **100** (2019) 066006 [1905.08261].
[83] H.-C. Kim, C. Vafa and K. Xu, *Finite landscape of 6d $N = (1, 0)$ supergravity*, *SciPost Phys.* **20** (2026) 016 [2411.19155].

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